

Exact analytical spectrum, eigenstates, and quantum geometry of the quarter-flux Harper-Hofstadter model

Isaac Tesfaye ^{1,*} and André Eckardt ^{1,†}

¹*Institut für Physik und Astronomie, Technische Universität Berlin, Hardenbergstr. 36, D-10623 Berlin, Germany*

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Quantum geometry has emerged as a guiding principle across atomic and condensed-matter physics, shaping the topological responses of Bloch bands and the stability of the correlated phases they host. Beyond two-band models, however, closed-form expressions for both the spectrum and the quantum geometry are rare. Here we provide such expressions for a paradigmatic four-band model that has recently been realized experimentally with ultracold atoms, photons and in superconducting circuits: the Harper-Hofstadter model at quarter flux, describing charged particles on a two-dimensional square lattice subjected to a uniform magnetic field. We achieve this by first showing that the model possesses a sublattice symmetry, which renders its Bloch Hamiltonian anti-block-diagonal allowing us to derive the spectrum and the eigenstates analytically. From that we also obtain closed-form expressions for the full quantum geometric tensor (QGT), including both the Berry curvature and the quantum metric, for all the bands of the model. For this purpose we first derive a general expression for the QGT for sublattice-symmetric systems in terms of contributions from the individual sublattice sectors. Finally, we evaluate fractional-Chern-insulator stability criteria analytically and quantify the lowest band of the quarter-flux Harper-Hofstadter model to be a nearly ideal Chern band.

Introduction – The quantum geometry of Bloch bands has emerged as a unifying language across atomic and condensed matter physics [1–7]. The quantum geometric tensor (QGT), whose imaginary part is the Berry curvature and whose real part is the quantum metric [1, 2, 8–13], governs topological responses such as the quantized Hall conductance [14–17], the superfluid weight of flat-band superconductors [18–24], orbital magnetism [25, 26], (nonlinear) optical responses [27–38], and the localization of Wannier functions [39–43]. Importantly, the QGT (which itself has also been generalized to other settings [44–57]) also governs a growing family of quantum-geometric bounds on observables [33, 58–73]. A prominent example of such a bound is related to the stabilization of fractional Chern insulators (FCIs) [74–80], which are lattice analogs of fractional quantum Hall states [81–84]. Such states have recently been realized in small systems of ultracold atoms [85–87] and photons [88, 89]. When a Chern band saturates this quantum-geometric bound, it is said to be ideal (or vortexable) [58–60, 90–96], as its quantum geometry then mimics that of the lowest Landau level (LLL) and becomes an especially favorable FCI host [58, 91, 94, 97].

Assessing these criteria typically requires the full quantum geometry of the Bloch band, yet closed-form expressions are scarce [98, 99]. For two-band models, the QGT follows from the Bloch vector, but Chern bands relevant for FCIs generically live in topological multiband settings where analytic solutions for both spectrum and quantum geometry are rare. A paradigmatic multiband system is the Harper-Hofstadter (HH) model [100–102], describing charged particles tunneling on a square lattice that is threaded by a uniform magnetic flux $\Phi = 2\pi\alpha$ per plaquette. This textbook example of a topologi-

cal band insulator with nonzero Chern numbers underlying the quantized Hall conductances [14, 17] has become a workhorse of quantum simulation, realized with ultracold atoms, both in two-dimensional optical lattices [103–106] and by combining a one-dimensional optical lattice with a synthetic dimension [107, 108], in photonic systems [109], and with superconducting qubits [110]. Despite its prominence, and although a rational flux $\alpha = p/q$ per plaquette (with p, q coprime) reduces the problem to a $q \times q$ Bloch Hamiltonian, a closed-form solution for the model has not been reported beyond the $q = 3$ three-band case [98, 99], for $q > 3$ both the spectrum and the quantum geometry have so far been accessible only numerically or perturbatively [99, 111].

In this work, we analytically solve the quarter-flux ($\alpha = 1/4$) Harper-Hofstadter model, obtaining closed-form expressions for the spectrum, the eigenstates, and the quantum geometry of all four bands as a function of quasimomentum $\mathbf{k}$. From the latter we also evaluate the momentum-based FCI-stability criteria analytically and show that the lowest Hofstadter band nearly approaches an ideal Chern band. The key observation of our work, allowing us to find closed-form expressions, is that among all possible choices of magnetic unit cell (MUC) for $\alpha = 1/4$, the symmetric 2×2 MUC uniquely preserves the sublattice symmetry of the model in quasimomentum space. This renders the 4×4 Bloch Hamiltonian anti-block-diagonal, reducing the problem to a singular-value decomposition of a remaining 2×2 block. Moreover, we show and use that for *any* sublattice-symmetric system, the Abelian and non-Abelian QGT decompose into contributions from the two sublattice sectors.

Sublattice symmetry in the quarter-flux Harper-Hofstadter model – We study the Harper-Hofstadter

Hamiltonian [100–102] in the Landau gauge

$$\hat{H} = -J \sum_{m,n} (e^{-i\Phi n} |m+1, n\rangle \langle m, n| + |m, n+1\rangle \langle m, n| + \text{h.c.}) \quad (1)$$

with a quarter flux per plaquette of $\Phi = 2\pi\alpha = \pi/2$ ($\alpha = 1/4$) and a hopping amplitude J . Here, $|m, n\rangle$ denotes the single-particle state localized at lattice site (m, n) , where m (n) denotes the discrete position along the x (y) direction of the square lattice of size $M \times N$, with M (N) the number of lattice sites along x (y).

Under periodic boundary conditions, the Hamiltonian (1) at rational flux $\alpha = p/q = 1/4$ admits a magnetic translation symmetry with respect to a *magnetic unit cell* (MUC) enclosing $q = 4$ plaquettes of total flux 2π [112–115]. Its area $A_{\text{MU}} = 4a^2$ is fixed (lattice constant $a = 1$ in the following), but its shape is not, giving three choices for $\alpha = 1/4$: (i) 1×4 , (ii) 2×2 , and (iii) 4×1 , each with four sublattice sites A, B, C, D , as shown in Fig. 1(a)–(b). A generalized Bloch theorem based on the associated magnetic translation operators (MTOs) yields, for each choice, a 4×4 Bloch Hamiltonian $H(\mathbf{k})$ acting on the sublattice basis states $|\mathbf{k}, \tau\rangle$, $\tau = A, B, C, D$, with bands $E_n(\mathbf{k})$, $n \in \{1, 2, 3, 4\}$, and quasimomentum $\mathbf{k} = (k_x, k_y)$. Different from the commonly studied case (i), we focus on case (ii) with the symmetric square MUC, for which the Bloch Hamiltonian reads [105]

$$H(\mathbf{k}) = -2J \begin{pmatrix} 0 & \cos(k_x a) & \cos(k_y a) & 0 \\ \cos(k_x a) & 0 & 0 & -\sin(k_y a) \\ \cos(k_y a) & 0 & 0 & -i \sin(k_x a) \\ 0 & -\sin(k_y a) & i \sin(k_x a) & 0 \end{pmatrix}. \quad (2)$$

The Bloch Hamiltonians for the other two choices (i) and (iii) of the MUC are given in Sec. S1 of the Supplemental Material [116]. We stress that all three choices of the MUC, and hence the Bloch Hamiltonians $H(\mathbf{k})$, originate from the same parent Hamiltonian (1) in a *fixed* gauge [117], are unitarily equivalent, and share the same energy spectrum as a function of $\mathbf{k}$ [116]. The first goal of this work is to find an exact analytical solution for the single-particle eigenstates and energies as a function of quasimomentum $\mathbf{k}$ by diagonalizing (2).

To this end, we utilize the chiral symmetry that is present for the Harper-Hofstadter model (1). A chiral symmetry is defined as a unitary involution $\hat{S}$ ($\hat{S}^2 = 1$) that anti-commutes with the Hamiltonian, $\{\hat{H}, \hat{S}\} = 0$, [118–122] and splits the Hilbert space $\mathcal{H} = \mathcal{H}_+ \oplus \mathcal{H}_-$ into two orthogonal subspaces $\mathcal{H}_{\pm}$ of dimension $d_{\pm} = \dim \mathcal{H}_{\pm}$ corresponding to the eigenvalues ± 1 of $\hat{S}$. In the eigenbasis of $\hat{S}$, $\hat{H}$ is anti-block-diagonal and its spectrum is symmetric about zero, with non-zero energies $E_{\pm, j} = \pm \sigma_j$ set by the singular values σ_j of the off-diagonal block D . The eigenstates of each chiral pair

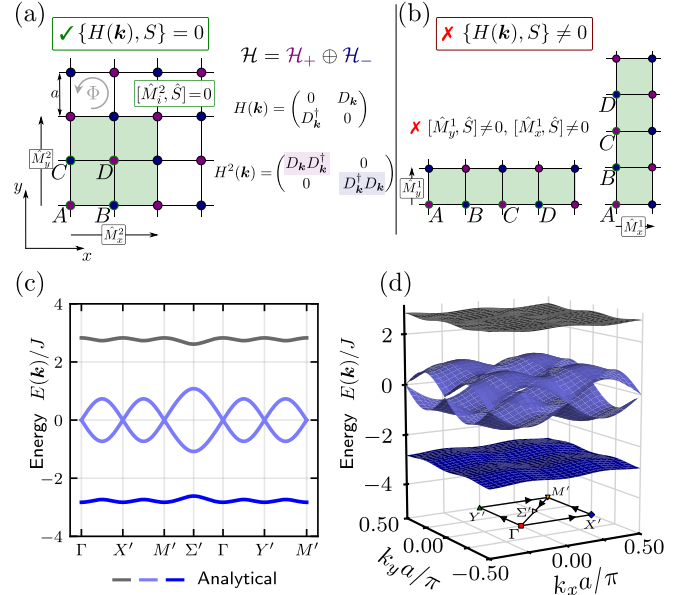


FIG. 1. Sublattice symmetry and analytical spectrum of the quarter-flux Harper-Hofstadter model. (a) Real space lattice of HH-model with a quarter flux $\Phi = \pi/2$ per plaquette, the symmetric 2×2 MUC (green shaded area) with its four sublattice sites A, B, C, D . The sublattice symmetry $\hat{S} = (-1)^{m+n} |m, n\rangle \langle m, n|$ corresponding to the bipartite checkerboard pattern (purple/blue colored sites for the even/odd sublattice), commutes with the associated magnetic translation operators (MTOs) $\hat{M}_i^2$ of the MUC, $[\hat{S}, \hat{M}_i^2] = 0$, and is preserved in momentum space via $S = \text{diag}(1, -1, -1, 1)$, and anti-commutes with the Bloch Hamiltonian (2), $\{H(\mathbf{k}), S\} = 0$. (b): Same as (a) for the other two choices of the MUC, (i) and (iii), of size 1×4 and 4×1 , respectively, showing that the sublattice symmetry is not preserved in momentum space as $\{H(\mathbf{k}), S\} \neq 0$ due to $[\hat{S}, \hat{M}_i^1] \neq 0$ [116]. (c) Energy spectrum $E_n(\mathbf{k})$ (5) along the high-symmetry path [defined in (d)] in the FBZ with solid lines indicate the analytical results. (d) $E_n(\mathbf{k})$ over the full FBZ.

always take the form [123]

$$|\Psi_{\epsilon, j}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} |u_j\rangle \\ \epsilon |v_j\rangle \end{pmatrix}, \quad (3)$$

where $|u_j\rangle \in \mathcal{H}_+$ ($|v_j\rangle \in \mathcal{H}_-$) are the left (right) singular vectors of D [124]. Here, $\epsilon = \pm 1$ labels the energy pair $E_{\pm, j}$ and $j = 1, \dots, r$ with $r = \text{rank}(D)$.

The chiral symmetry results from the bipartite checkerboard pattern $s_{m, n} = (-1)^{m+n}$ of the real-space square lattice, which allows us to divide the lattice sites into even (odd) sites with $s_{m, n} = +1$ ($s_{m, n} = -1$), as illustrated in Fig. 1(a)–(b) via the purple (blue-colored) lattice sites, respectively. Thus, for $\hat{H}$ (1) we have $\{\hat{H}, \hat{S}\} = 0$, with $\hat{S} = \sum_{m, n} s_{m, n} |m, n\rangle \langle m, n|$, because there are only inter-sublattice couplings (between even and odd sites), but no intra-sublattice couplings (between even and even or odd and odd sites).

Now, the key idea of our work is realizing that, when going to momentum space, the distinction between a chiral and a sublattice symmetry becomes crucial. Namely, a sublattice symmetry is a chiral symmetry whose operator is diagonal in the site basis, assigning a fixed sign $s = \pm 1$ to every site and thereby partitioning the lattice into two sign-definite sublattices and providing a *grading* [116]. In real space the checkerboard operator $\hat{S}$ is both, since its even-odd pattern $s_{m,n}$ is itself such a grading, with the “even” $s = 1$ sublattice containing states with $\tau = A$ and D and the “odd” $s = -1$ sublattice containing states with $\tau = B$ and C [Fig. 1(a)]. In momentum space, however, it is only the choice (ii) of the MUC with the Bloch Hamiltonian $H(\mathbf{k})$ (2) for which the chiral operator remains diagonal in the sublattice basis $|\mathbf{k}, \tau\rangle$ of (2), and hence preserves the sublattice symmetry. Indeed, in (2) we see that there are no coupling terms between even sublattice sites (A and D) or odd sublattice sites (B and C), but only coupling terms connecting even with odd sublattice sites and vice versa [Fig. 1(a)]. We stress that this is *not* the case for the other two choices of the MUC (i) and (iii), whose associated magnetic translation operators (MTOs) include a single-site translations that do not commute with the real-space sublattice symmetry operator $\hat{S}$ [Fig. 1(b)], so that $\hat{S}$ is not preserved as a grading in momentum space. In these cases $H(\mathbf{k})$ contains diagonal terms, such that no even-odd grading of the four sublattices τ can anti-commute with it [116].

With time-reversal and particle-hole symmetry absent and a $\mathbf{k}$ -independent chiral symmetry operator present, the model (2) belongs to the chiral class AIII of the Altland-Zirnbauer classification [125, 126], for which no winding-number invariant should be present in two dimensions [123]. This is consistent with the absence of an energy gap at zero energy [Fig. 1(c)-(d)], making the ground state of non-interacting fermions topologically trivial at Fermi energy of $E_F = 0$. However, when only the lowest band is occupied, the Fermi energy E_F takes a negative value in gap above the lowest band, which breaks the chiral symmetry. This places the model into symmetry class A, which allows for a topological Chern invariant in even spatial dimensions, consistent with non-zero Chern number of the lowest band [116].

Analytical spectrum of the quarter-flux Harper-Hofstadter model – From the illustration in Fig. 1(a), we see that the sublattice sites A and D (B and C) within the 2×2 MUC form the even (odd) sublattice sector, such that sublattice symmetry operator in its eigenbasis reads $S = \text{diag}(+1, +1, -1, -1)$. A site permutation of (2) brings it to the anti-block-diagonal form $H_S(\mathbf{k}) = \begin{pmatrix} 0 & D_{\mathbf{k}} \\ D_{\mathbf{k}}^\dagger & 0 \end{pmatrix}$, with the 2×2 block

$$D_{\mathbf{k}} = -2J \begin{pmatrix} \cos(k_x a) & \cos(k_y a) \\ -\sin(k_y a) & i \sin(k_x a) \end{pmatrix}, \quad (4)$$

mapping the odd to the even sublattice sector. Squar-

ing gives $H_S^2 = \text{diag}(D_{\mathbf{k}} D_{\mathbf{k}}^\dagger, D_{\mathbf{k}}^\dagger D_{\mathbf{k}})$, so the energies are the square roots of the eigenvalues of the two Hermitian blocks, $M_{\mathbf{k}} \equiv D_{\mathbf{k}}^\dagger D_{\mathbf{k}}$ and $\tilde{M}_{\mathbf{k}} \equiv D_{\mathbf{k}} D_{\mathbf{k}}^\dagger$. Their non-zero eigenvalues, with eigenvectors $|v_j\rangle$ and $|u_j\rangle$, respectively, are the same and coincide with the squared singular values σ_j^2 of $D_{\mathbf{k}} = U_{\mathbf{k}} \Sigma_{\mathbf{k}} V_{\mathbf{k}}^\dagger = \sum_{j=1}^{r=2} \sigma_j |u_j\rangle \langle v_j|$ [124]. We write $M_{\mathbf{k}} = D_{\mathbf{k}}^\dagger D_{\mathbf{k}} = h_0 \sigma_0 + \mathbf{h} \cdot \boldsymbol{\sigma}$ in Pauli form with $h_0 = 4J^2$ and, in the shorthand notation $c_{i\nu} \equiv \cos(ik_\nu a)$, $s_{i\nu} \equiv \sin(ik_\nu a)$, the Bloch vector $\mathbf{h} = (h_x, h_y, h_z) = 4J^2(c_x c_y, s_x s_y, \frac{1}{2}(c_{2x} - c_{2y}))$ (omitting the $\mathbf{k}$ -dependence for brevity). Its magnitude is $h \equiv |\mathbf{h}| = J^2 \sqrt{2Z}$ with $Z = 6 + c_{4x} + c_{4y}$, so the eigenvalues are $\lambda_j = h_0 + \eta_j |\mathbf{h}|$ with $\eta_j = (-1)^{j+1}$ ($j \in \{1, 2\}$). Taking square roots, the four energy bands $E_{\pm,j} = \pm \sqrt{\lambda_j}$ of model follow as

$$E_{\pm,j}(\mathbf{k})/J = \pm \sqrt{4 + \eta_j \sqrt{2} \sqrt{6 + \cos(4k_x a) + \cos(4k_y a)}}, \quad (5)$$

which is the first main result of this work [116]. In Fig. 1(c)-(d) we show the exact analytical energy spectrum (5) along a high-symmetry path in the first magnetic Brillouin zone (FBZ) [Fig. 1(c)] as well as over the entire FBZ (d). Ordering the spectrum from lowest ($n = 1$) to highest ($n = 4$) energy, we have the four bands $E_n(\mathbf{k}) \in \{E_{-,1}, E_{-,2}, E_{+,2}, E_{+,1}\}$. The middle two bands $E_{-,2}$ and $E_{+,2}$ touch at certain high-symmetry points in the FBZ, forming Dirac cones, and hence should actually be thought of as a single *super-band* [105]. From the sublattice symmetry it follows that the spectrum is symmetric around zero, $E_{-,j}(\mathbf{k}) = -E_{+,j}(\mathbf{k})$ (5) with $j \in \{1, 2\}$.

Having obtained the exact energy spectrum, let us now compute the eigenstates. To this end, note that the right singular vectors $|v_j\rangle$ of $D_{\mathbf{k}}$ are also the eigenvectors of $M_{\mathbf{k}}$, hence fixed by the Bloch vector $\mathbf{h}$, and the left singular vectors follow from the SVD relation $|u_j\rangle = D_{\mathbf{k}} |v_j\rangle / \sigma_j$ with $\sigma_j = E_{+,j} > 0$ (5). Note that this SVD relation is crucial as it fixes the relative phase between $|v_j\rangle$ and $|u_j\rangle$, and is what makes $|\Psi_{\epsilon,j}\rangle$ (3) an eigenstate of $H_S(\mathbf{k})$. This gives the four energy eigenstates $|\Psi_{\epsilon,j}\rangle = (|u_j\rangle, \epsilon |v_j\rangle)^T / \sqrt{2}$ (3) for the four bands $E_{\epsilon,j} = \epsilon E_{+,j}$ ($\epsilon = \pm 1, j \in \{1, 2\}$) (5). In particular, the ground state of the lowest band $E_{-,1}$, the state typically populated in cold-atom realizations of the quarter-flux Harper-Hofstadter model [103–105], is $|\Psi_{-,1}\rangle = (|u_1\rangle, -|v_1\rangle)^T / \sqrt{2}$. The explicit closed-form of $|v_j\rangle, |u_j\rangle$, an equivalent Bloch-sphere representation in the polar and azimuthal angles $(\theta_{\mathbf{k}}, \varphi_{\mathbf{k}})$ of $\hat{\mathbf{h}} = \mathbf{h}/h$, and all four eigenstates are given in the [End Matter](#).

QGT for sublattice-symmetric systems – The sublattice structure that solves the spectrum also allows us to obtain the full quantum-geometrical characterization for the model in closed form. We find that for any sublattice-symmetric system, whose eigenstates take the form of (3), the Abelian and the non-Abelian quantum

geometric tensor (QGT) [2, 12] decompose into contributions from the two sublattice sectors. In particular, the Abelian QGT $Q_{\mu\nu}^n = \langle \partial_\mu \Psi_n | (1 - |\Psi_n\rangle\langle\Psi_n|) | \partial_\nu \Psi_n \rangle = g_{\mu\nu}^n - \frac{i}{2} \Omega_{\mu\nu}^n$ of an isolated band $n \equiv (\epsilon, j)$, always takes the form

$$Q_{\mu\nu}^j = \frac{1}{2}(Q_{\mu\nu}^{(u_j)} + Q_{\mu\nu}^{(v_j)}) + \frac{1}{4} \Delta A_\mu^j \Delta A_\nu^j, \quad (6)$$

where $Q^{(u_j)}$ ($Q^{(v_j)}$) is the QGT of the singular vector $|u_j\rangle$ ($|v_j\rangle$), and $\Delta A_\mu^j \equiv A_\mu^{(u_j)} - A_\mu^{(v_j)}$ is the difference of their Berry connections $A_\mu^{(w)} = i\langle w | \partial_\mu w \rangle$ with $w \in \{u_j, v_j\}$ [127]. Note that Eq. (6) is independent of ϵ , any two chiral partner bands $E_{\pm,j}$ (that are gapped) always share the same quantum geometry. Splitting it into its real and symmetric part, the quantum metric $g_{\mu\nu}^j = \Re[Q_{\mu\nu}^j]$, and its imaginary and antisymmetric part, the Berry curvature $\Omega_{\mu\nu}^j = -2\Im[Q_{\mu\nu}^j]$, while using that $\Delta A_\mu^j \Delta A_\nu^j$ is purely real and symmetric, we obtain

$$\begin{aligned} g_{\mu\nu}^j &= \frac{1}{2}(g_{\mu\nu}^{(u_j)} + g_{\mu\nu}^{(v_j)}) + \frac{1}{4} \Delta A_\mu^j \Delta A_\nu^j, \\ \Omega_{\mu\nu}^j &= \frac{1}{2}(\Omega_{\mu\nu}^{(u_j)} + \Omega_{\mu\nu}^{(v_j)}). \end{aligned} \quad (7)$$

We find that the connection differences enter the metric only, while the Berry curvature is simply given by the sector average. Thus, both members of a chiral pair carry identical Berry curvature, $\Omega_{\mu\nu}^j \equiv \Omega_{\mu\nu}^{(+,j)} = \Omega_{\mu\nu}^{(-,j)}$, and therefore identical Chern numbers, $C_{+,j} = C_{-,j}$.

Application to the HH-model – Here, each sector is a two-level Bloch problem set by $M = D^\dagger D$ and $\tilde{M} = D D^\dagger$ and their Bloch vectors $\mathbf{h}, \tilde{\mathbf{h}}$, so the sector QGTs and hence the full $Q_{\mu\nu}^j$ (6) can be obtained analytically [116]. For the quarter-flux ground band this yields the analytical expression for the Berry curvature

$$\Omega_{xy}^1(\mathbf{k}) = -\sqrt{2} a^2 \frac{6 - \cos(4k_x a) - \cos(4k_y a)}{[6 + \cos(4k_x a) + \cos(4k_y a)]^{3/2}}, \quad (8)$$

together with the analytical expressions for all three quantum metric components $g_{\mu\nu}^1(\mathbf{k})$ (see [End Matter](#)), which are all also equal to that of the highest band $E_{+,1}$. Since $\Omega_{xy}^1 < 0$ throughout the FBZ, the curvature never changes sign, consistent with the Chern number $C_1 \equiv C_{-,1} = \int_{\text{BZ}} (\Omega_{xy}^1 / 2\pi) d^2k = -1$ of the lowest Hofstadter band, which, due to (7) coincides with that of its chiral partner (highest band), $C_4 \equiv C_{+,1} = C_1 = -1$.

The degenerate middle super-band is instead governed by the non-Abelian QGT [12, 128], which we find to admit an analogous sublattice decomposition (see the [End Matter](#) and [116]). For its non-Abelian Berry curvature F_{xy} , we independently obtain $\text{tr} F_{xy} = -2\Omega_{xy}^1$ (8), consistent with the conservation law that the total Berry curvature summed over all bands vanishes [17], and fixing the super-bands' total Chern number to $C_{2+3} = 2$.

A nearly ideal Chern band – Several band-geometric criteria quantify a band's stability to host fractional

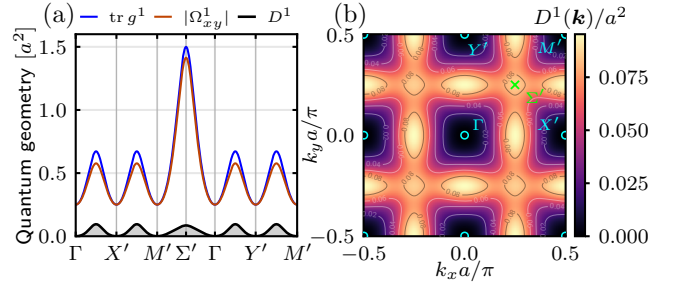


FIG. 2. Local trace-condition (ideal-band) defect $D^1(\mathbf{k}) = \text{tr} g^1(\mathbf{k}) - |\Omega_{xy}^1(\mathbf{k})| \geq 0$ (9) of the quarter-flux Harper-Hofstadter ground band $|\Psi_{-,1}\rangle$ obtained from the analytical Berry curvature (8) and the quantum metric (7). (a) $D^1(\mathbf{k})$ along the high-symmetry path Γ - X' - M' - Σ' - Γ - Y' - M' [as in Fig. 1], together with $\text{tr} g^1$ (blue) and $|\Omega_{xy}^1|$ (red) overlaid, which fluctuate across the FBZ, but coincide at the high-symmetry points (Γ , X' , Y' , M' cyan circles in (b)), so the shaded gap D^1 (black) stays small. (b) $D^1(\mathbf{k})$ depicted over the 2D FBZ. The defect is pointwise non-negative and vanishes exactly at the cyan-colored high-symmetry points where the band is locally an ideal LLL while it peaks near the midpoint of Γ - X' (and its C_4 equivalent). Its integral over the entire Brillouin zone equals the global trace-condition $\mathcal{T}_{\text{HH}}$ (9).

Chern insulators (FCIs), most of them based on lowest-Landau-level (LLL) “mimicry”. These are momentum-based conditions reproducing the QGT of the LLL [58–60, 79, 90, 94, 97, 99, 129, 130], or, more recently, real-space conditions such as vortexability, aimed at platforms without a simple momentum-space description [91–93]. We test these criteria on the lowest Chern band using our exact QGT. Positive semi-definiteness of the QGT yields a hierarchy of local inequalities $\text{tr} g(\mathbf{k}) \geq 2\sqrt{\det g(\mathbf{k})} \geq |\Omega_{xy}(\mathbf{k})|$ in 2D [18, 58–60]. A band saturating the local trace defect $D^1(\mathbf{k}) = \text{tr} g^1(\mathbf{k}) - |\Omega_{xy}^1(\mathbf{k})| \geq 0$ everywhere is a so-called ideal (vortexable) Chern band on the lattice [60, 91, 97]. The integrated version of this trace defect, the (global) trace-condition [91],

$$\mathcal{T}_{\text{HH}} = \int_{\text{BZ}} D^1(\mathbf{k}) d^2k \geq 0, \quad (9)$$

is a single number that summarizes the deviation from ideality. Saturating it with uniform curvature further leads to the famous Girvin-MacDonald-Platzman algebra [131] with Landau-level form factors [58, 90, 132, 133]. The lowest band of the quarter-flux Harper-Hofstadter model realizes this approximately [116]: using the exact curvature (8) and exact metric (7) gives us $D^1(\mathbf{k})$ in closed form, which we depict in Fig. 2. Although the ground band is not geometrically flat, with its Berry curvature and metric fluctuating across the Brillouin zone [Fig. 2(b)], it falls short of saturating the global trace condition (9) by only $\approx 8.3\%$ of the topological bound $2\pi|C_1|$ [61, 62], so that the band is “nearly ideal” in the integrated sense [116].

Note that this defect is also directly measurable: $\mathcal{T}_{\text{HH}}$ controls the small- $\mathbf{q}$ static structure factor, $S(\mathbf{q}) \sim K_{\mu\nu} q^\mu q^\nu + \mathcal{O}(q^4)$ through the so-called quantum weight $K_{\mu\nu} = (2\pi)^{-1} \int_{\text{BZ}} g_{\mu\nu} d^2k$ [61, 62], which is isotropic here with $K \equiv K_{xx} + K_{yy} = |C_1| + \mathcal{T}_{\text{HH}}/2\pi \approx 1.083 > |C_1| = 1$, exceeding the topological bound $K \geq |C_1| = 1$ [61, 62] (see Sec. S7 of [116] for details).

This near-ideality is complemented by the flatness of the band itself, a typical requirement for FCI stability, since it appears favorable when interactions dominate the kinetic energy scale [74, 80, 134] [135]. From the spectrum (5) the exact bandwidth and gap are $W/J = 2\sqrt{2} - \sqrt{4 + 2\sqrt{2}}$ and $\Delta/J = 2\sqrt{2} - \sqrt{2}$, giving the exact flatness ratio $f = \Delta/W \approx 7.1097$ [136]. Without fine tuning this is flatter than the Haldane model [137], which at its optimal fine-tuned point has a flatness ratio of $f \approx 6.0136$ [74, 138, 139]. Models with even flatter (or exactly flat) bands exist [74, 77, 130, 134, 140–147], but they rely on (fine-tuned) longer-range hoppings, such as the exponentially decaying couplings of the Kapit-Mueller model [145], or on fine-tuned system sizes at which the Hofstadter bands themselves become exactly flat [148], all of which remain experimentally challenging to realize. The quarter-flux Harper-Hofstadter model, by contrast, needs only nearest-neighbor tunneling and has already been realized on several platforms [103–110].

Summary – We have analytically solved the quarter-flux Harper-Hofstadter model by exploiting its sublattice symmetry that survives in quasimomentum space only for a certain symmetric magnetic unit cell. The resulting anti-block-diagonal Bloch Hamiltonian reduces the four-band problem to a single 2×2 block, yielding the spectrum and all four eigenstates analytically. We then showed that for any such sublattice-symmetric system the quantum geometric tensor decomposes into contributions from the two sublattice sectors, giving us analytical expressions for the Berry curvature and quantum metric of the gapped bands, and their non-Abelian counterparts for the degenerate middle pair. For the ground band this exposes an energetically flat Chern band that is nearly FCI-ideal, which we quantified via the quantum-geometric trace condition. As our QGT decomposition is completely general and holds for any sublattice-symmetric Bloch Hamiltonian, we envision that it can be applied to other multi-band systems, which, provided the off-diagonal block D is small enough, may even be obtained in closed form.

Within the Hofstadter family with $\alpha = p/q$ flux per plaquette this is restrictive as the sublattice symmetry survives in momentum space only for q divisible by four (and $q = 8$ already requires solving a 4×4 block). Thus, non-trivial N -band candidates ($N = d_+ + d_- > 2$) are more natural elsewhere. For instance, with balanced sublattices ($d_+ = d_-$) the BBH quadrupole insulator [149, 150] and the square-octagon lattice [151–

153] are four-site bipartite models, where D is a 2×2 block. For a majority sublattice ($d_+ \neq d_-$), the dice and Lieb lattices [19, 118] as well as other chiral flat-band networks [121] are interesting candidates to study. Here, D is rectangular with $|d_+ - d_-|$ zero modes, so the Abelian QGT decomposition can handle the dispersive bands while the degenerate flat sector requires the non-Abelian counterpart [154]. When larger blocks resist closed-form diagonalization, it might also still be interesting to combine the decomposition with eigenprojector methods for N -band quantum geometry [155].

Finally, our analytical results might help to understand, why the Harper-Hofstadter model at quarter flux hosts a bosonic Laughlin-like FCI ground state at half filling of the lowest band, even in the limit of infinitely strong on-site interactions [156].

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* i.tesfaye@tu-berlin.de

† eckardt@tu-berlin.de

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- [176] We use the Fourier convention $e^{-i\mathbf{q} \cdot \hat{\mathbf{r}}}$ throughout, which is opposite to that of [133].

End Matter

Exact eigenstates – The anti-block form (2) gives us all four eigenstates in closed form. With $M_{\mathbf{k}} = D_{\mathbf{k}}^\dagger D_{\mathbf{k}} = h_0 \sigma_0 + \mathbf{h} \cdot \boldsymbol{\sigma}$ (main text), the right singular vectors $|v_j\rangle$ of $D_{\mathbf{k}}$ are also eigenvectors of $M_{\mathbf{k}}$,

$$|v_j\rangle = \frac{1}{\sqrt{2h(h + \eta_j h_z)}} \begin{pmatrix} h + \eta_j h_z \\ \eta_j(h_x + i h_y) \end{pmatrix}, \quad (10)$$

with $\eta_j = (-1)^{j+1}$, satisfying $M_{\mathbf{k}}|v_j\rangle = \sigma_j^2|v_j\rangle$, while the left singular vectors follow from the phase-locked SVD relation $|u_j\rangle = D_{\mathbf{k}}|v_j\rangle/E_{+,j}$. With Eq. (3) this yields the four eigenstates $|\Psi_{\epsilon,j}\rangle = (|u_j\rangle, \epsilon|v_j\rangle)^T/\sqrt{2}$, which are explicitly given by

$$|\Psi_{\epsilon,j}\rangle = \begin{pmatrix} -\frac{2J}{E_{+,j}} [c_x(h + \eta_j h_z) + \eta_j c_y(h_x + i h_y)] \\ -\frac{2J}{E_{+,j}} [-s_y(h + \eta_j h_z) + i \eta_j s_x(h_x + i h_y)] \\ \epsilon(h + \eta_j h_z) \\ \epsilon \eta_j(h_x + i h_y) \end{pmatrix} \times \frac{1}{\sqrt{2} \sqrt{2h(h + \eta_j h_z)}} \quad (11)$$

in particular the ground state of the lowest band ($j = 1$, $\eta_1 = +1$) is given by $|\Psi_{-,1}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} |u_1\rangle \\ -|v_1\rangle \end{pmatrix}$ (see Sec. S4 of [116] for more details).

Sublattice decomposition of the quantum geometry – For any sublattice-symmetric band $n \equiv (\epsilon, j)$ whose eigenstate $|\Psi_{\epsilon,j}\rangle$ is given by Eq. (3), it follows that the Abelian quantum geometric tensor $Q_{\mu\nu}^n = \langle \partial_\mu \Psi_n | \partial_\nu \Psi_n \rangle - A_\mu^n A_\nu^n$, using for the Berry connection $2A_\mu^j = (A_\mu^{(u_j)} + A_\mu^{(v_j)})$ together with $2\langle \partial_\mu \Psi_{\epsilon,j} | \partial_\nu \Psi_{\epsilon,j} \rangle = \langle \partial_\mu u_j | \partial_\nu u_j \rangle + \langle \partial_\mu v_j | \partial_\nu v_j \rangle$, decomposes into the two sublattice-sector contributions via $Q_{\mu\nu}^j = \frac{1}{2}(Q_{\mu\nu}^{(u_j)} + Q_{\mu\nu}^{(v_j)}) + \frac{1}{4}\Delta A_\mu^j \Delta A_\nu^j$ (6). Since each sector is a two-level Bloch problem set by $\mathbf{h}, \tilde{\mathbf{h}}$ for $M_{\mathbf{k}}, \tilde{M}_{\mathbf{k}}$ respectively, the sector QGTs can be obtained via two-level quantum geometry machinery [155, 166], such that the full QGT of all isolated bands follows in closed form. For the lowest band $(\epsilon, j) = (-1, 1)$, $|\Psi_{-,1}\rangle$, this gives, the exact Berry curvature $\Omega_{xy}^1(\mathbf{k})$ (8), and the exact quantum metric components

$$\begin{aligned} g_{xx}^1/a^2 &= \frac{Z(2 - c_{4x}) - s_{4x}^2}{Z^2} + \frac{s_{2y}^2 c_{2x}^2}{\mathcal{G}_1^2}, \\ g_{yy}^1/a^2 &= \frac{Z(2 - c_{4y}) - s_{4y}^2}{Z^2} + \frac{s_{2x}^2 c_{2y}^2}{\mathcal{G}_1^2}, \\ g_{xy}^1/a^2 &= -\frac{s_{4x} s_{4y}}{Z^2} - \frac{s_{2x} s_{2y} c_{2x} c_{2y}}{\mathcal{G}_1^2}, \end{aligned} \quad (12)$$

with $\mathcal{G}_1 = 2\sqrt{2Z} + Z$. Thanks to the ϵ -independence of the right-hand side of Eq. (6), these are simultaneously the quantum metric components of the highest band $|\Psi_{+,1}\rangle$.

For the degenerate case the non-Abelian QGT $\mathcal{Q}_{\mu\nu}^{ab} = \langle \partial_\mu \Psi_a | \partial_\nu \Psi_b \rangle - \sum_c A_\mu^{ac} A_\nu^{cb} = G_{\mu\nu}^{ab} - \frac{i}{2} F_{\mu\nu}^{ab}$ [12, 167] becomes relevant, which for two pair of degenerate bands $a \equiv (\epsilon', j')$ and $b \equiv (\epsilon, j)$, obeys an analogous decomposition

$$\mathcal{Q}_{\mu\nu}^{(\epsilon', j'), (\epsilon, j)} = \frac{1}{2} [\mathcal{Q}_{\mu\nu}^{(u_{j'}, u_j)} + \epsilon' \epsilon \mathcal{Q}_{\mu\nu}^{(v_{j'}, v_j)}]. \quad (13)$$

Here, $\mathcal{Q}^{(u_{j'}, u_j)} = \langle \partial_\mu u_{j'} | \partial_\nu u_j \rangle - \sum_{j''} A_\mu^{(u_{j'}, u_{j''})} A_\nu^{(u_{j''}, u_j)}$ is the non-Abelian QGT of the left singular vectors $|u_j\rangle$, with the non-Abelian (interband) Berry connection [168] $A_\mu^{(u_{j'}, u_j)} = i \langle u_{j'} | \partial_\mu u_j \rangle$, and similarly for the right singular vectors $|v_j\rangle$. For the degenerate middle-band pair with $(\epsilon', j') = (+, 2)$ and $(\epsilon, j) = (-, 2)$, the non-Abelian QGT $\mathcal{Q}_{\mu\nu}^{(\epsilon', 2), (\epsilon, 2)}$ becomes a 2×2 matrix in the degenerate subspace with the trace of non-Abelian Berry curvature given by

$$\begin{aligned} \text{tr } F_{xy}/a^2 &= 2\sqrt{2} \frac{6 - \cos(4k_x a) - \cos(4k_y a)}{[6 + \cos(4k_x a) + \cos(4k_y a)]^{3/2}} \\ &= -2 \Omega_{xy}^1. \end{aligned} \quad (14)$$

This fixes the middle super-band's total Chern number to $C_{2+3} \equiv C_{(-,2), (+,2)} = (2\pi)^{-1} \int_{\text{BZ}} \text{tr } F_{xy} d^2 k = -2 C_1 = 2$, while the trace of the non-Abelian metric components $\text{tr } G_{\mu\nu}$ are given by

$$\begin{aligned} \text{tr}(G_{xx})/a^2 &= \frac{2[Z(2 - c_{4x}) - s_{4x}^2]}{Z^2}, \\ \text{tr}(G_{yy})/a^2 &= \frac{2[Z(2 - c_{4y}) - s_{4y}^2]}{Z^2}, \\ \text{tr}(G_{xy})/a^2 &= \frac{-2s_{4x}s_{4y}}{Z^2}. \end{aligned} \quad (15)$$

The full details of the derivations are given in Sec. S5 of [116].

Supplemental Material for “Exact analytical spectrum, eigenstates, and quantum geometry of the quarter-flux Harper-Hofstadter model”

Isaac Tesfaye ^{1,*} and André Eckardt ^{1,†}

¹*Institut für Physik und Astronomie, Technische Universität Berlin, Hardenbergstr. 36, D-10623 Berlin, Germany*
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CONTENTS

S1. Magnetic translation symmetry in the Harper-Hofstadter model	1
A. Quarter-flux case	2
S2. Chiral and sublattice symmetry	3
A. Sublattice symmetry, class A/AIII, and C_4 in the quarter-flux Harper-Hofstadter model	4
S3. Details on analytical solution of the quarter-flux Harper-Hofstadter model	6
S4. Details on the eigenstates of the quarter-flux Harper-Hofstadter model for the symmetric $2a \times 2a$ magnetic unit cell (MUC)	6
S5. Quantum Geometry of sublattice symmetric systems.	8
A. Non-Abelian Quantum Geometry of sublattice symmetric systems	11
S6. FCI Stability Conditions	14
A. Momentum-space hierarchy: from local positivity to integrated conditions	14
B. Application to the quarter-flux Harper-Hofstadter ground band	15
C. Relation to the projected density algebra	16
S7. Static structure factor and the quantum weight	17

This Supplemental Material is organized as follows. In Section [S1](#) we review the magnetic translation symmetry and the generalized Bloch theorem for the three magnetic-unit-cell choices, specialized to the quarter-flux case. Section [S2](#) provides more details regarding the sublattice symmetry, establishing the class AIII assignment, and briefly discusses the C_4 symmetry of the quarter-flux model. In Section [S3](#) we provide more details on the SVD-based derivation of the exact spectrum, after which in Section [S4](#) the four closed-form eigenstates of the symmetric $2a \times 2a$ MUC are derived. Then in Section [S5](#) we derive the quantum geometry of sublattice-symmetric systems, including the non-Abelian quantum geometric tensor of the degenerate middle bands. Section [S6](#) establishes the hierarchy of FCI stability conditions and applies it to the lowest Chern band. Finally, in Section [S7](#) we briefly relate the static structure factor to the quantum weight and its lower bound.

S1. MAGNETIC TRANSLATION SYMMETRY IN THE HARPER-HOFSTADTER MODEL

In this section, we briefly review the magnetic translation symmetry [[112](#), [113](#)] of the Harper-Hofstadter model for the specific case of a quarter flux per plaquette. For a more extensive discussion on magnetic translation symmetries, we refer to Refs. [[114](#), [115](#)]. We consider the Harper-Hofstadter Hamiltonian [[100](#), [102](#)] in the Landau gauge $(\phi_{m,n}^x, \phi_{m,n}^y) = (-\Phi n, 0)$ given by

$$\begin{aligned}\hat{H} &= -J \sum_{m,n} (e^{i\phi_{m,n}^x} |m+1, n\rangle \langle m, n| + e^{i\phi_{m,n}^y} |m, n+1\rangle \langle m, n| + \text{h.c.}) \\ &= -J \sum_{m,n} (e^{-i\Phi n} |m+1, n\rangle \langle m, n| + |m, n+1\rangle \langle m, n| + \text{h.c.}),\end{aligned}\tag{S1}$$

with a flux per plaquette of $\Phi = 2\pi\alpha$ and a hopping amplitude J . Here $|m, n\rangle$ denotes the single-particle state localized at lattice site (m, n) , where m (n) denotes the discrete position along the x (y) direction of the square lattice of size $M \times N$, with M (N) the number of lattice sites along x (y). We work in the single-particle sector throughout, so the results are independent of particle

statistics.

Then, for periodic boundary conditions (PBC), the Hamiltonian ([S1](#)) for rational values of flux per plaquette, $\alpha = p/q$ with $p, q \in \mathbb{Z}$ being coprime integers, admits a magnetic translation symmetry with respect to a *magnetic unit cell* (MUC) of size $k \times \ell$ with $k\ell = q \times \mathbb{Z}$ [[112](#), [113](#)]. In other words, one can construct

a full set of commuting operators if one can find a region of dimension $k \times \ell$, called *supercell*, which encloses multiple plaquettes whose total flux is an integer multiple of 2π (magnetic flux quantum). The magnetic unit cell is then defined as the supercell with the smallest dimension which satisfies this condition,

$$k\ell\Phi = k\ell 2\pi\alpha = 2\pi p \frac{k\ell}{q} \stackrel{!}{=} 2\pi\lambda, \lambda \in \mathbb{Z}, \quad (\text{S2})$$

where the solution is given by $k\ell = q$ [114, 115]. Note that only the total area of the magnetic unit cell, $A_{\text{MU}} = (k \times \ell)a^2$ is fixed, where the specific dimension is not. Due to the enlarged magnetic unit cell of size $k \times \ell$ with $k\ell = q$, we obtain a reduced first (magnetic) Brillouin zone of size, $0 \leq k_x < 2\pi/(ka)$ (or $-\pi/(ka) \leq k_x < \pi/(ka)$) and $0 \leq k_y < 2\pi/(\ell a)$ (or $-\pi/(\ell a) \leq k_y < \pi/(\ell a)$), with $\mathbf{k} = (k_x, k_y)$ being the quasimomenta of the system and a the lattice spacing constant. Because of the enlarged magnetic unit cell the allowed quasimomenta for the first (magnetic) Brillouin zone, for a system of size $M \times N$ with PBC, are given by $k_x = 2\pi\alpha_x/(Ma)$ and $k_y = 2\pi\alpha_y/(Na)$, where $\alpha_x \in \{0, 1, \dots, M/k - 1\}$ and $\alpha_y \in \{0, 1, \dots, N/\ell - 1\}$ are integers and where k (ℓ) is the dimension of the MUC along x (y). For simplicity, we will assume in the following, that M (N) are always integer multiple of k (ℓ).

In the case of the Landau gauge specified above, one can find the full set of commuting operators $\{\hat{H}, \hat{M}_x^k, \hat{M}_y^\ell\}$ where the so-called commuting magnetic translation operators (MTOs) $\hat{M}_x^k$ and $\hat{M}_y^\ell$ are given by [114, 115]

$$\begin{aligned} \hat{M}_x^k &= \sum_{m,n} |m+k, n\rangle \langle m, n|, \\ \hat{M}_y^\ell &= \sum_{m,n} e^{-i2\pi\alpha\ell m} |m, n+\ell\rangle \langle m, n|. \end{aligned} \quad (\text{S3})$$

Then the simultaneous single-particle eigenstates $|\psi\rangle = \sum_{m,n} \psi_{m,n} |m, n\rangle$ of the set of commuting operators $\{\hat{H}, \hat{M}_x^k, \hat{M}_y^\ell\}$ are obtained by using the *generalized Bloch theorem* based on the MTOs $\hat{M}_x^k$ and $\hat{M}_y^\ell$

$$\begin{aligned} \langle m, n | \hat{M}_x^k | \psi \rangle &= e^{i\mu_{m,n}^x(k)} \psi_{m-k, n} = e^{-ik_x ka} \psi_{m, n}, \\ \langle m, n | \hat{M}_y^\ell | \psi \rangle &= e^{i\mu_{m,n}^y(\ell)} \psi_{m, n-\ell} = e^{-ik_y \ell a} \psi_{m, n}, \end{aligned} \quad (\text{S4})$$

where $\mu_{m,n}^x(k)$ ($\mu_{m,n}^y(\ell)$) denote the phases arising from the application of operators $\hat{M}_x^k$ ($\hat{M}_y^\ell$) [114, 115].

A. Quarter-flux case

Hence, for the specific case of a quarter flux per plaquette, $\alpha = 1/4$, there are three different choices for the magnetic unit cell (MUC), namely (i) $A_{\text{MU}} = ka \times \ell a = 1a \times 4a$, (ii) $A_{\text{MU}} = 2a \times 2a$ and (iii) $A_{\text{MU}} = 4a \times 1a$, as

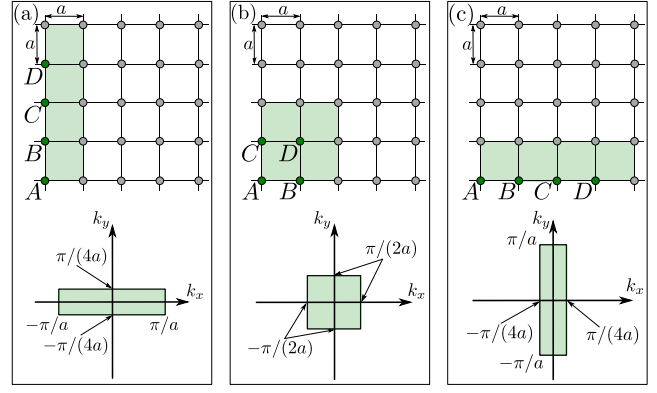


FIG. S1. Admissible magnetic unit cells (MUC) and first magnetic Brillouin zones (FBZ) for a square lattice with $\alpha = 1/4$ flux per plaquette. Three different possibilities for the choice of the MUC are available. Each MUC contains necessarily $q = 4$ lattice sites (dark green circles), labeled by the letters A, B, C, D . (a) [Choice (i)] Rectangular choice for the magnetic unit cell along y enclosing an area of $A_{\text{MU}} = 1a \times 4a$ (green shaded area) with an associated FBZ of size $-\pi/a \leq k_x < \pi/a$ and $-\pi/(4a) \leq k_y < \pi/(4a)$ in quasimomentum space. (b) [Choice (ii)] Symmetric square-shaped choice for the MUC, $A_{\text{MU}} = 2a \times 2a$ leading to a FBZ of size $-\pi/(2a) \leq k_i < \pi/(2a)$ for $i \in \{x, y\}$. (c) [Choice (iii)] Rectangular choice for the magnetic unit cell along x covering an area of $A_{\text{MU}} = 4a \times 1a$ with a corresponding FBZ of extent $-\pi/(4a) \leq k_x < \pi/(4a)$ and $-\pi/a \leq k_y < \pi/a$.

illustrated in Fig. S1. Each MUC consists of four sublattice sites A, B, C, D and four plaquettes enclosing a total area of $A_{\text{MU}} = 4a^2$, which is pierced by a total flux of $4 \times 2\pi\alpha = 2\pi$, as required by the condition above (S2).

Using a generalized Bloch theorem based on the magnetic translation operators associated with the MUC [114, 115], we can construct the single-particle quasimomentum Bloch Hamiltonian $H(\mathbf{k})$ associated to the eigenstates $|\mathbf{k}, \mu\rangle = \sum_{m,n} \phi_{m,n}^{\mathbf{k}, \mu} |m, n\rangle$ for the three different choices of the MUC, which we label as cases (i), (ii) and (iii). Here, $\mathbf{k} = (k_x, k_y)$ is the quasimomentum of the system and $\mu \in \{1, 2, 3, 4\}$ denotes band index for the system. Since each MUC contains four sublattice sites, the Bloch Hamiltonian $H(\mathbf{k})$ is a 4×4 matrix acting on the so-called Bloch function $\mathbf{u}_\mu(\mathbf{k}) = (u_\mu^A(\mathbf{k}), u_\mu^B(\mathbf{k}), u_\mu^C(\mathbf{k}), u_\mu^D(\mathbf{k}))^T$, with the eigenvalue equation $H(\mathbf{k})\mathbf{u}_\mu(\mathbf{k}) = E_\mu(\mathbf{k})\mathbf{u}_\mu(\mathbf{k})$.

For our purposes the case of (ii) is the relevant one because it is the only case for which the sublattice symmetry of the quarter-flux Harper-Hofstadter model manifests itself, which we discuss in more detail now. In the case (ii) of the symmetric MUC, we have $k = \ell = 2$ such

that MTOs $\hat{M}_x^2$ and $\hat{M}_y^2$ of (S3) take the form

$$\begin{aligned}\hat{M}_x^2 &= \sum_{m,n} |m+2, n\rangle\langle m, n|, \\ \hat{M}_y^2 &= \sum_{m,n} e^{-i\pi m} |m, n+2\rangle\langle m, n|.\end{aligned}\quad (\text{S5})$$

The phase factor in $\hat{M}_y^2$ somewhat complicates the conditions imposed by the generalized Bloch theorem (S4). We find the single-particle eigenstates of the system in this case, $|\psi\rangle = \sum_{m,n} \psi_{m,n} |m, n\rangle$, by proposing an ansatz for the coefficients $\psi_{m,n}$, which must satisfy the generalized Bloch theorem (S4)

$$\begin{aligned}\langle m, n | \hat{M}_x^2 | \psi \rangle &= \psi_{m-2, n} = e^{-ik_x 2a} \psi_{m, n}, \\ \langle m, n | \hat{M}_y^2 | \psi \rangle &= e^{-i\pi m} \psi_{m, n-2} = e^{-ik_y 2a} \psi_{m, n}.\end{aligned}\quad (\text{S6})$$

An Ansatz which satisfies these conditions (S6) is given by [105]

$$\psi_{m,n} = e^{ik_x m a} e^{ik_y n a} \times \begin{cases} u^A, & \text{for } m, n \text{ even,} \\ u^B e^{i\pi n/2}, & \text{for } m \text{ odd, } n \text{ even,} \\ u^C, & \text{for } m \text{ even, } n \text{ odd,} \\ u^D e^{i\pi n/2}, & \text{for } m, n \text{ odd,} \end{cases} \quad (\text{S7})$$

where u^i denote the complex amplitudes of the Bloch functions on the sublattice sites $i \in \{A, B, C, D\}$ in each MUC, as shown in Fig. S1(b). Inserting this ansatz into the eigenvalue equation $H|\psi\rangle = E|\psi\rangle$, we can obtain the Bloch Hamiltonian $H(\mathbf{k})$ for case (ii) given by

$$\begin{aligned}H^{(\text{ii})}(\mathbf{k}) &= \\ &-2J \begin{pmatrix} 0 & \cos(k_x a) & \cos(k_y a) & 0 \\ \cos(k_x a) & 0 & 0 & -\sin(k_y a) \\ \cos(k_y a) & 0 & 0 & -i \sin(k_x a) \\ 0 & -\sin(k_y a) & i \sin(k_x a) & 0 \end{pmatrix},\end{aligned}\quad (\text{S8})$$

For completeness, let us also briefly discuss the Bloch Hamiltonians $H(\mathbf{k})$ for case (i) and case (iii). In the commonly studied case of vertically aligned rectangular MUC (i) with $A_{\text{MU}} = 1a \times 4a$ [Fig. S1(a)] there are no phase factors in either of the MTOs $\hat{M}_x^1$ and $\hat{M}_y^4$. This simplifies the conditions imposed by the generalized Bloch theorem (S4) such that a simple plane-wave ansatz, $\psi_{m,n} = e^{ik_x m a} e^{ik_y n a} u^n$ with $u^{n-4} = u^n$, suffices for the coefficients of the single-particle eigenstate in this case [114, 115]. It follows that the Bloch Hamiltonian for case (i) is then given by

$$H^{(\text{i})}(\mathbf{k}) = -J \begin{pmatrix} h_1 & e^{ik_y a} & 0 & e^{-ik_y a} \\ e^{-ik_y a} & h_2 & e^{ik_y a} & 0 \\ 0 & e^{-ik_y a} & h_3 & e^{ik_y a} \\ e^{ik_y a} & 0 & e^{-ik_y a} & h_4 \end{pmatrix}, \quad (\text{S9})$$

where $h_j = 2 \cos(k_x a + (\pi/2)j)$ for $j \in \{1, 2, 3, 4\}$.

For case (iii), we have that $\hat{M}_y^1$ carries a phase factor of $e^{-i\pi m/2}$, so that the ansatz, $\psi_{m,n} = e^{ik_x m a} e^{ik_y n a} e^{-i\pi m n/2} u^m$ with $u^{m-4} = u^m$, leads to the following Bloch Hamiltonian

$$H^{(\text{iii})}(\mathbf{k}) = -J \begin{pmatrix} g_1 & e^{ik_x a} & 0 & e^{-ik_x a} \\ e^{-ik_x a} & g_2 & e^{ik_x a} & 0 \\ 0 & e^{-ik_x a} & g_3 & e^{ik_x a} \\ e^{ik_x a} & 0 & e^{-ik_x a} & g_4 \end{pmatrix} \quad (\text{S10})$$

where $g_j = 2 \cos(k_y a - (\pi/2)j)$ for $j \in \{1, 2, 3, 4\}$.

We stress that the three MUC choices describe the same physics and do *not* correspond to different gauges (e.g. Landau vs symmetric gauge). Choosing a MUC is choosing how to group the lattice into supercells for the generalized Bloch theorem (S4), a labeling choice made independently of the gauge, since the admissibility condition of integer flux through the supercell (S2) is gauge invariant. Fixing the single Landau-gauge Hamiltonian (S1), all three MUCs supply commuting MTOs for that *one* $\hat{H}$ (S1), so the three 4×4 Bloch Hamiltonians are the same operator written in three magnetic-Bloch bases, with eigenstates that are the same physical states relabeled.

S2. CHIRAL AND SUBLATTICE SYMMETRY

Here, we collect some general properties of finite-dimensional chiral-symmetric systems that are used in the main text, after which we discuss their application to the three MUC choices of the quarter-flux Harper-Hofstadter model. Since the two terms, *chiral* and *sublattice* symmetry, are often used interchangeably, we clarify the distinction in the following.

A *chiral symmetry* is a unitary involution $\hat{S}$ (with $\hat{S}^2 = \mathbb{1}$, and hence $\hat{S} = \hat{S}^{-1} = \hat{S}^\dagger$) that anti-commutes with the Hamiltonian, $\{\hat{H}, \hat{S}\} = 0$ [118–122, 157–159],

$$\hat{S}^2 = \mathbb{1}, \quad \{\hat{H}, \hat{S}\} = 0. \quad (\text{S11})$$

In the sense of the tenfold way [125] only these two properties (S11), if present, place a Hamiltonian in class AIII for instance.

Now, a *sublattice symmetry* is the stronger requirement that the involution is also a *grading* of the lattice, i.e., that $\hat{S}$ be diagonal in the local site basis $|\tau\rangle$ with entries ± 1 ,

$$\hat{S} = \sum_{\tau} s_{\tau} |\tau\rangle\langle\tau|, \quad s_{\tau} = \pm 1, \quad (\text{S12})$$

so that each site carries a definite sign and the eigenspaces $\mathcal{H}_{\pm}$ are spanned by sets of sites rather by arbitrary superpositions of them. Every grading is a chiral operator, while the converse is not true. We stress

that the grading, and not chirality alone, is what the following sections use. It gives the off-diagonal block D introduced below the meaning of an inter-sublattice hopping matrix, and it makes a decomposition into the $\mathcal{H}_\pm$ sectors a decomposition by sublattice sites.

For completeness, below we list some general properties of *chirally*-symmetric systems, which are used throughout this work, which all follow the two properties (S11) alone.

Eigenspace splitting. As a unitary involution $\hat{S}$ has eigenvalues ± 1 , so the Hilbert space splits into its eigenspaces, $\mathcal{H} = \mathcal{H}_+ \oplus \mathcal{H}_-$ with $d_\pm = \dim \mathcal{H}_\pm$ so that in its eigenbasis $\hat{S} = \text{diag}(\mathbb{1}_{d_+}, -\mathbb{1}_{d_-})$.

Anti-block-diagonal form. In the $\hat{S}$ -eigenbasis the Hamiltonian is purely off-diagonal, $\hat{H} = \begin{pmatrix} 0 & D \\ D^\dagger & 0 \end{pmatrix}$ with $D : \mathcal{H}_- \rightarrow \mathcal{H}_+$ and vice versa $D^\dagger : \mathcal{H}_+ \rightarrow \mathcal{H}_-$. The diagonal blocks are forced to vanish by $\{\hat{H}, \hat{S}\} = 0$ together with $\hat{S} = \text{diag}(\mathbb{1}, -\mathbb{1})$.

Spectral symmetry. The spectrum is symmetric about zero. That means if $\hat{H}|\psi_E\rangle = E|\psi_E\rangle$ then $\hat{H}(\hat{S}|\psi_E\rangle) = -E(\hat{S}|\psi_E\rangle)$, such that $\hat{S}|\psi_E\rangle$ is an eigenstate with energy $-E$.

Singular-value spectrum. Squaring the Hamiltonian gives $\hat{H}^2 = \text{diag}(DD^\dagger, D^\dagger D)$, whose non-zero eigenvalues are the squared singular values σ_j^2 of D . Hence, the non-zero energies of $\hat{H}$ come in chiral pairs $E_{\pm,j} = \pm\sigma_j$ with $j = 1, \dots, r$ and $r = \text{rank}(D)$. This follows from the SVD, $D = U\Sigma V^\dagger = \sum_{j=1}^r \sigma_j |u_j\rangle\langle v_j|$ and the fact that $DD^\dagger$ and $D^\dagger D$ share their non-zero spectrum [124].

Paired eigenstates. The eigenstates of a pair $\pm\sigma_j$ always take the form of (3)

$$|\Psi_{\epsilon,j}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} |u_j\rangle \\ \epsilon |v_j\rangle \end{pmatrix} \quad (\text{S13})$$

with $\epsilon = \pm 1$, where $|u_j\rangle \in \mathcal{H}_+$ and $|v_j\rangle \in \mathcal{H}_-$ are the left and right singular vectors of D , respectively. It follows from inserting the SVD relations $D|v_j\rangle = \sigma_j|u_j\rangle$ and $D^\dagger|u_j\rangle = \sigma_j|v_j\rangle$ into the energy eigenvalue equation $\hat{H}|\Psi_{\epsilon,j}\rangle = \epsilon\sigma_j|\Psi_{\epsilon,j}\rangle$ (in the chiral basis).

Protected zero modes. The dimension of the kernel of $\hat{H}$ (the zero-energy subspace) is $d_0 = \dim \ker \hat{H} = |d_+ - d_-| + 2(\min(d_+, d_-) - r) \geq |d_+ - d_-|$, and the zero modes $|\psi_0\rangle$ can be chosen within a single chiral sector, i.e., $|\psi_0\rangle \in \ker D^\dagger \subset \mathcal{H}_+$ or $\ker D \subset \mathcal{H}_-$, following from the rank-nullity theorem applied to D [118, 119, 121, 159]. Whenever $\hat{S}$ is a (grading) sublattice symmetry (S12), these sectors are sublattices and the zero modes are supported on one sublattice only.

A. Sublattice symmetry, class A/AIII, and C_4 in the quarter-flux Harper-Hofstadter model

We now apply these properties to the quarter-flux Harper-Hofstadter model, where the bipartite checker-

board sign $s_{m,n} = (-1)^{m+n}$ defines the real-space chiral operator $\hat{S} = \sum_{m,n} s_{m,n} |m,n\rangle\langle m,n|$. The Hamiltonian (S1) anti-commutes with $\hat{S}$, since the square lattice is bipartite with neither diagonal couplings nor on-site potentials, and in real space $\hat{S}$ is manifestly a grading (S12), being diagonal in the site basis with entries $s_{m,n} = \pm 1$. The two notions coincide there, so that the question is whether this coincidence survives in quasimomentum space.

When $\hat{S}$ commutes with the magnetic translations of the chosen cell, the two can be simultaneously diagonalized, so that $\hat{S}$ descends to a $\mathbf{k}$ -independent diagonal matrix in the cell basis $|\mathbf{k}, \tau\rangle$ and $\{H(\mathbf{k}), S\} = 0$ at fixed $\mathbf{k}$. This holds for the symmetric cell (ii), where $[\hat{S}, \hat{M}_x^2] = [\hat{S}, \hat{M}_y^2] = 0$. For the rectangular cells the single-site translations instead anti-commute, $\{\hat{S}, \hat{M}_x^1\} = 0$ for (i) and $\{\hat{S}, \hat{M}_y^1\} = 0$ for (iii), thereby exchanging the even-odd sublattice labels [Fig. 1(a)-(b)]. The operators themselves are collected in Section S1, and the two cases are the class-I versus class-II distinction mentioned in Ref. [122].

The consequence for the Bloch Hamiltonians follows from splitting the checkerboard sign into its two factors, $s_{m,n} = (-1)^m(-1)^n$. In the symmetric cell both factors act internally, so that $\hat{S}$ descends to the $\mathbf{k}$ -independent grading $S^{(\text{ii})} = \text{diag}(1, -1, -1, 1)$ with

$$S^{(\text{ii})} H^{(\text{ii})}(\mathbf{k}) S^{(\text{ii})} = -H^{(\text{ii})}(\mathbf{k}), \quad (\text{S14})$$

so the spectrum is reflected about zero energy at each *fixed* $\mathbf{k}$. For the rectangular cells only $(-1)^n$ stays internal, becoming $-S_\pi$, while $(-1)^m = e^{i\pi m}$ acts, after the partial Fourier transform along x , as a momentum shift $k_x \rightarrow k_x + \pi/a$. The chiral relation therefore acquires that shift,

$$S_\pi H^{(\text{i})}(k_x, k_y) S_\pi = -H^{(\text{i})}(k_x + \pi/a, k_y), \quad (\text{S15})$$

where $S_\pi = \text{diag}(1, -1, 1, -1)$ written in the sublattice site basis of the case-(i) ansatz (S9), and analogously $S_\pi H^{(\text{iii})}(k_x, k_y) S_\pi = -H^{(\text{iii})}(k_x, k_y + \pi/a)$ for case (iii). The energies (S28) then satisfy $E(k_x, k_y) = -E(k_x + \pi/a, k_y)$ in case (i) and $E(k_x, k_y) = -E(k_x, k_y + \pi/a)$ in case (iii), so the chiral partner of a state at $\mathbf{k}$ is displaced by half a reciprocal lattice vector.

A diagonal $S = \text{diag}(s_j)$ would give $\{H, S\}_{jj} = 2s_j H_{jj}$, which vanishes only if every diagonal entry does, while $H^{(\text{i})}(\mathbf{k})$ (S9) carries the on-site terms $-Jh_j$ with $h_j = 2\cos(k_x a + (\pi/2)j)$. Note that these h_j are $\mathbf{k}$ -dependent, antiperiodic kinetic terms ($h_{j+2} = -h_j$), not a static on-site potential, and are fully compatible with chirality, as we will see below. Contrary to this, a global chemical potential $\mu \mathbb{1}$ is a different matter, as required to ensure quarter-filling for instance, since $\mathbb{1}$ is chiral-even, $\{H - \mu \mathbb{1}, S\} = -2\mu S \neq 0$ for *every* chiral involution S , so a nonzero μ breaks the chiral symmetry and moves

the model from class AIII to class A, as discussed in the main text. The lowest-band Chern number is accordingly an ordinary class-A invariant, a property of the isolated band protected by its gap. Thus, for cases (i) and (iii) a $\mathbf{k}$ -independent chiral operator nevertheless exists,

$$S^{(i)} = S^{(iii)} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \quad (\text{S16})$$

which is Hermitian and unitary with $(S^{(i)})^2 = \mathbb{1}$ and $\{H^{(i)}(\mathbf{k}), S^{(i)}\} = 0$ at fixed $\mathbf{k}$, and likewise for case (iii) with (S10). By being off-diagonal it is only a chiral operator mapping site j to $j + 2$. There is therefore no contradiction with $\{\hat{S}, \hat{M}_x^1\} = 0$, since $S^{(i)}$ is the Bloch representative not of $\hat{S}$ but of $\hat{S}' = \hat{S}\hat{M}_y^2$, which still anti-commutes with $\hat{H}$, i.e., $\{\hat{H}, \hat{S}'\} = 0$. However, $\hat{S}'$ now commutes with both magnetic translations of cell (i), $[\hat{S}', \hat{M}_x^1] = [\hat{S}', \hat{M}_y^1] = 0$, because $\hat{M}_y^2$ anti-commutes with $\hat{M}_x^1$ just as $\hat{S}$ does.

The three magnetic-Bloch bases are related by $\mathbf{k}$ -independent unitaries, which are given by

$$WH^{(i)}(\mathbf{k})W^\dagger = H^{(ii)}(\mathbf{k}), \quad W = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 & 1 \\ 0 & -1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ -i & 0 & i & 0 \end{pmatrix}, \quad (\text{S17})$$

with columns labeled by the sublattice sites $j = 1, \dots, 4$ of cell (i) and rows by $\{A, B, C, D\}$ of cell (ii).

Note that the ± 1 eigenvectors of $S^{(i)}$ are the superpositions $(|1\rangle \pm |3\rangle)/\sqrt{2}$ and $(|2\rangle \mp |4\rangle)/\sqrt{2}$ of the cell-(i) sites. By rotating $H^{(i)}$ and $H^{(iii)}$ into this $S^{(i)}$ -eigenbases we can also bring them to the anti-block-diagonal form $\begin{pmatrix} 0 & D \\ D^\dagger & 0 \end{pmatrix}$ as well, with the off-diagonal blocks

$$D_{\mathbf{k}}^{(i)} = \begin{pmatrix} 2J \sin(k_x a) & -2J \cos(k_y a) \\ 2iJ \sin(k_y a) & 2J \cos(k_x a) \end{pmatrix}, \quad (\text{S18})$$

$$D_{\mathbf{k}}^{(iii)} = \begin{pmatrix} -2J \sin(k_y a) & -2J \cos(k_x a) \\ 2iJ \sin(k_x a) & 2J \cos(k_y a) \end{pmatrix}. \quad (\text{S19})$$

These are *not* equal to $D_{\mathbf{k}}^{(ii)}$ (S22), but they share its singular values: in all three cases $\text{tr}(D^\dagger D) = 8J^2$ and $\det(D^\dagger D) = 2J^4 [2 - \cos(4k_x a) - \cos(4k_y a)]$, so that they reproduce the same spectrum (S28) as expected. What our exact solution relies on is the *grading*, namely, only in case (ii) does the anti-block form live on the physical sublattice basis. There, $D_{\mathbf{k}}^{(ii)}$ is a genuine inter-sublattice hopping matrix, its singular vectors are site-basis Bloch functions (S13), and the sublattice decomposition of the quantum geometry becomes a decomposition into the two *physical* sublattices. The abstract superposition sectors of cases (i) and (iii) provide none of this, even though their quantum geometry itself is identical in all three

MUC descriptions, as it must be for a gauge-invariant observable. To summarize, the grading of case (ii) provides two things instead. First, it gives the closed-form eigenstates (S13), from which the quantum geometry follows analytically. Second, it decomposes that geometry into the two sublattice sectors $\mathcal{H}_\pm$, which is physically meaningful because those sectors are spanned by lattice sites (see Section S5).

With time-reversal and particle-hole symmetry absent and a $\mathbf{k}$ -independent chiral operator present, all three descriptions of $H(\mathbf{k})$ belong to the chiral class AIII of the tenfold Altland-Zirnbauer classification [125, 126], as they must, being three magnetic-Bloch bases of one and the same Hamiltonian which dictates that in $d = 2$ dimension, at half-filling at zero energy, there is no topological invariant associated to the band below that gap [123]. This is consistent with the fact that the HH-model at quarter-flux does not feature any isolated band at half-filling as it is gapless at zero energy there. Introducing a chemical potential $\mu \neq 0$ to ensure quarter-filled lowest band breaks the chiral symmetry and moves the model to class A, where the lowest band is isolated and carries a nonzero Chern number $C_1 = -1$ consistent with AZ-class A in $d = 2$.

Let us briefly discuss the point-group symmetry of the square lattice in the presence of the flux. The bare rotation $\mathcal{P} = \sum_{m,n} | -n, m \rangle \langle m, n |$ maps the Landau gauge along x into the Landau gauge along y , and is therefore not a symmetry by itself. The symmetry is the composite $\mathcal{C}_4 = \mathcal{G}\mathcal{P}$ with the compensating gauge phase $\chi_{m,n} = -\Phi mn$ (assuming for simplicity a linear system size of a multiple of $q = 4$). Restricting $e^{-i\Phi mn}$ to the four sites of the symmetric cell gives $\text{diag}(1, 1, 1, i)$, since mn is odd only at $D = (a, a)$, while $\mathcal{P}$ acts within the cell as the transposition Π_{BC} , so that we have

$$R^{(ii)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & i \end{pmatrix},$$

$$R^{(ii)} H^{(ii)}(k_x, k_y) R^{(ii)\dagger} = H^{(ii)}(-k_y, k_x), \quad (\text{S20})$$

with $(R^{(ii)})^4 = \mathbb{1}$, so that C_4 symmetry is represented linearly. The rotation commutes with the sublattice operator, $[R^{(ii)}, S^{(ii)}] = 0$, so that C_4 acts as an ordinary crystalline symmetry on top of the chiral class AIII or class A (if a chemical potential is added). We caution that, in analogy with the chiral operator discussed above, the existence of such a constant R does *not* single out the symmetric cell. Namely, the intertwining condition (S17) allows for a similar construction also for the MUCs associated to cases (i) and (iii). Since $WH^{(iii)}(k_x, k_y)W^\dagger = H^{(ii)}(-k_y, k_x)$, the unitary $W^{(iii)} = R^{(ii)\dagger}W$ relates case (iii) to case (ii) at the same quasimomentum.

We mention in passing that the C_4 symmetry endows

the Chern bands with a concrete, quantized secondary invariant, the *discrete shift* $\mathcal{S}$ as detailed in [160, 161]. The discrete shift is the lattice analog of the Wen-Zee shift, defined for $U(1)$ -charge-conserving fermions with C_M rotational symmetry and valued in $\mathbb{Z}_M$. For C_4 ($M = 4$) and spinless fermions the crystalline response theory fixes $\mathcal{S} \equiv C/2 \pmod{1}$, so that a band with *odd* Chern number necessarily carries a *half-integer* shift [162, 163], so $\mathcal{S} = -1/2 \pmod{1}$ for the lowest band of the quarter-flux Harper-Hofstadter model here. Physically, $\mathcal{S}$ governs a curvature/defect response distinct from the chiral edge mode: a $\pi/2$ disclination centered on a C_4 -invariant site binds a fractional charge $\tilde{Q} = \mathcal{S}/4 = 1/8 \pmod{1}$ on top of the background filling $\nu_0 = 1/4$ [162, 163].

S3. DETAILS ON ANALYTICAL SOLUTION OF THE QUARTER-FLUX HARPER-HOFSTADTER MODEL

We collect some of the algebraic details behind the exact solution summarized in the main text, which will also be used later for the derivation of the eigenstates (Sec. S4) and the exact quantum geometry (Sec. S5). For the symmetric $2a \times 2a$ MUC (ii), the bipartite checkerboard sign $s_{m,n} = (-1)^{m+n}$ (see Section S2) assigns the sites A, D to the even sector $\mathcal{H}_+ = \text{span}\{|A\rangle, |D\rangle\}$ and B, C to the odd sector $\mathcal{H}_- = \text{span}\{|B\rangle, |C\rangle\}$, so that the sublattice operator in its eigenbasis is $S = \text{diag}(\mathbb{1}_2, -\mathbb{1}_2)$. Rotating $H^{(\text{ii})}(\mathbf{k})$ (2) into the eigenbasis $\{|A\rangle, |D\rangle, |B\rangle, |C\rangle\}$ by the permutation Q (with $Q|A\rangle = |A\rangle$, $Q|B\rangle = |D\rangle$, $Q|C\rangle = |B\rangle$, $Q|D\rangle = |C\rangle$) brings it to the anti-block-diagonal form enforced by the sublattice symmetry,

$$H_S^{(\text{ii})}(\mathbf{k}) = Q^\dagger H^{(\text{ii})}(\mathbf{k}) Q = \begin{pmatrix} 0 & D_{\mathbf{k}} \\ D_{\mathbf{k}}^\dagger & 0 \end{pmatrix}, \quad (\text{S21})$$

with the 2×2 block $D_{\mathbf{k}} : \mathcal{H}_- \rightarrow \mathcal{H}_+$ given by

$$D_{\mathbf{k}} = -2J \begin{pmatrix} \cos(k_x a) & \cos(k_y a) \\ -\sin(k_y a) & i \sin(k_x a) \end{pmatrix}, \quad (\text{S22})$$

as (4) in the main text. Squaring $H_S^{(\text{ii})}$ (S21) block-diagonalizes it into the two Hermitian 2×2 blocks $\tilde{M}_{\mathbf{k}} = D_{\mathbf{k}} D_{\mathbf{k}}^\dagger$ (acting on $\mathcal{H}_+$) and $M_{\mathbf{k}} = D_{\mathbf{k}}^\dagger D_{\mathbf{k}}$ (acting on $\mathcal{H}_-$),

$$(H_S^{(\text{ii})}(\mathbf{k}))^2 = \begin{pmatrix} \tilde{M}_{\mathbf{k}} & 0 \\ 0 & M_{\mathbf{k}} \end{pmatrix}. \quad (\text{S23})$$

Both admit a Pauli decomposition $M_{\mathbf{k}} = h_0 \sigma_0 + \mathbf{h} \cdot \boldsymbol{\sigma}$ and $\tilde{M}_{\mathbf{k}} = h_0 \sigma_0 + \tilde{\mathbf{h}} \cdot \boldsymbol{\sigma}$, with $h_\mu = \text{Tr}[M_{\mathbf{k}} \sigma_\mu]/2$, $\tilde{h}_\mu = \text{Tr}[\tilde{M}_{\mathbf{k}} \sigma_\mu]/2$, $h_0 = \tilde{h}_0 = 4J^2$, (dropping the explicit $\mathbf{k}$ -dependence from now on) and the Bloch vectors

$$\mathbf{h} = 4J^2 \begin{pmatrix} c_x c_y \\ s_x s_y \\ \frac{1}{2}(c_{2x} - c_{2y}) \end{pmatrix}, \quad \tilde{\mathbf{h}} = 4J^2 \begin{pmatrix} -c_x s_y \\ s_x c_y \\ \frac{1}{2}(c_{2x} + c_{2y}) \end{pmatrix}, \quad (\text{S24})$$

with the same shorthand $c_{i\nu} = \cos(ik_\nu a)$, $s_{i\nu} = \sin(ik_\nu a)$ ($\nu \in \{x, y\}$, $i \in \mathbb{N}^+$) as in the main text. The two Bloch vectors have equal magnitude, $h \equiv |\mathbf{h}| = |\tilde{\mathbf{h}}| \equiv \tilde{h} = J^2 \sqrt{2Z}$ with $Z = 6 + c_{4x} + c_{4y}$, reflecting that M and $\tilde{M}$ share their non-zero eigenvalues [124],

$$\lambda_j = h_0 + \eta_j h, \quad \lambda_{1,2}/J^2 = 4 \pm \sqrt{2} \sqrt{6 + c_{4x} + c_{4y}}, \quad (\text{S25})$$

with the eigenvalue selector $\eta_j = (-1)^{j+1} \in \{+1, -1\}$ (not to be confused with the energy sign $\epsilon = \pm 1$ of the chirally symmetric pairs). These are the squared singular values $\lambda_j = \sigma_j^2$ of the singular value decomposition of $D_{\mathbf{k}}$,

$$D_{\mathbf{k}} = U_{\mathbf{k}} \Sigma_{\mathbf{k}} V_{\mathbf{k}}^\dagger = \sum_{j=1}^r \sigma_j |u_j\rangle \langle v_j|, \quad (\text{S26})$$

with rank $r = \text{rank}(D_{\mathbf{k}}) = 2$ and $\sigma_j \equiv E_{+,j} > 0$, from which

$$M = D^\dagger D = V \Sigma^\dagger \Sigma V^\dagger = \sum_{j=1}^r \sigma_j^2 |v_j\rangle \langle v_j|, \quad \tilde{M} = D D^\dagger = U \Sigma \Sigma^\dagger U^\dagger = \sum_{j=1}^r \sigma_j^2 |u_j\rangle \langle u_j|. \quad (\text{S27})$$

Hence the right and left singular vectors $|v_j\rangle$ and $|u_j\rangle$ are simultaneously the eigenvectors of M and $\tilde{M}$ with eigenvalue $\lambda_j = \sigma_j^2$. Taking square roots, the four bands $E_{\pm,j} = \pm \sqrt{\lambda_j}$ read

$$E_{\pm,1}(\mathbf{k})/J = \pm \sqrt{4 + \sqrt{2} \sqrt{6 + \cos(4k_x a) + \cos(4k_y a)}}, \quad E_{\pm,2}(\mathbf{k})/J = \pm \sqrt{4 - \sqrt{2} \sqrt{6 + \cos(4k_x a) + \cos(4k_y a)}}, \quad (\text{S28})$$

reproducing the main-text spectrum (5).

S4. DETAILS ON THE EIGENSTATES OF THE QUARTER-FLUX HARPER-HOFSTADTER MODEL FOR THE SYMMETRIC $2a \times 2a$ MAGNETIC UNIT CELL (MUC)

Having obtained the exact energy spectrum (S28), we now construct all four eigenstates $|\Psi_{\epsilon,j}\rangle$ of $H_S(\mathbf{k})$ (S21) in closed form. Recall from Sec. S3 and (S27) that the eigenvectors $|v_j\rangle$ of M corresponding to the eigenvalues λ_j are simultaneously the right singular vectors of D (S26). Expressed directly through the Bloch-vector components in the (north-pole) gauge, $|v_j\rangle$ reads [164, 165]

$$|v_j\rangle = \frac{1}{\sqrt{2h(h + \eta_j h_z)}} \begin{pmatrix} h + \eta_j h_z \\ \eta_j (h_x + i h_y) \end{pmatrix}, \quad (\text{S29})$$

which is the gauge used throughout the quantum-geometry analysis as well. Equivalently, parametrizing the unit vector $\hat{\mathbf{h}} = \mathbf{h}/h = (\cos \varphi_{\mathbf{k}} \sin \theta_{\mathbf{k}}, \sin \varphi_{\mathbf{k}} \sin \theta_{\mathbf{k}}, \cos \theta_{\mathbf{k}})^T$ by its polar and azimuthal angles $(\theta_{\mathbf{k}}, \varphi_{\mathbf{k}})$, the same states take the Bloch-sphere form $|v_1\rangle = \cos(\theta_{\mathbf{k}}/2)|A\rangle + \sin(\theta_{\mathbf{k}}/2)e^{i\varphi_{\mathbf{k}}}|B\rangle$ and

$|v_2\rangle = \sin(\theta_{\mathbf{k}}/2)|A\rangle - \cos(\theta_{\mathbf{k}}/2)e^{i\varphi_{\mathbf{k}}}|B\rangle$, located at $\pm\hat{\mathbf{h}}$ on the Bloch sphere, where $|A\rangle$ ($|B\rangle$) denotes the north (south) pole.

The left singular vectors then follow from the SVD relation $|u_j\rangle = D_{\mathbf{k}}|v_j\rangle/\sigma_j$ (S26). Using $D_{\mathbf{k}} = -2J \begin{pmatrix} c_x & c_y \\ -s_y & is_x \end{pmatrix}$ (S22), this yields

$$|u_j\rangle = \frac{-2J}{\sigma_j \sqrt{2h(h + \eta_j h_z)}} \begin{pmatrix} c_x(h + \eta_j h_z) + \eta_j c_y(h_x + ih_y) \\ -s_y(h + \eta_j h_z) + i\eta_j s_x(h_x + ih_y) \end{pmatrix}. \quad (\text{S30})$$

We emphasize that $|u_j\rangle$ must be taken from this SVD relation, and *not* by independently gauge-fixing an eigenvector $|u_j\rangle$ of $\tilde{M}$ (S27). Although $|u_j\rangle$ is such an eigenvector (with Bloch vector $\tilde{\mathbf{h}}$, see Sec. S3), only the relative u/v phase, correctly fixed by $D|v_j\rangle = \sigma_j|u_j\rangle$, makes $(|u_j\rangle, \epsilon|v_j\rangle)^T$ an eigenstate of H_S . An independent gauge choice would break this fixed phase relationship and differ by a $\mathbf{k}$ -dependent phase. This is the same phase-locking mechanism that renders the Berry-connection difference ΔA_{μ}^j (S49) gauge invariant in the following quantum-geometry analysis of Section S5.

With $|v_j\rangle$ (S29) and $|u_j\rangle$ (S30) at hand, the four eigenstates $|\Psi_{\epsilon,j}\rangle$ of H_S (S21) follow from the general sublattice-symmetric form of (S13), with $\epsilon = \pm 1$ and $j \in \{1, 2\}$, so that (ϵ, j) enumerates the four bands $E_{\epsilon,j} = \epsilon E_{+,j}$ (S28). Inserting (S29) and (S30) gives the explicit closed form

$$|\Psi_{\epsilon,j}\rangle = \frac{1}{\sqrt{2} \sqrt{2h(h + \eta_j h_z)}} \begin{pmatrix} -\frac{2J}{E_{+,j}} [c_x(h + \eta_j h_z) + \eta_j c_y(h_x + ih_y)] \\ -\frac{2J}{E_{+,j}} [-s_y(h + \eta_j h_z) + i\eta_j s_x(h_x + ih_y)] \\ \epsilon(h + \eta_j h_z) \\ \epsilon\eta_j(h_x + ih_y) \end{pmatrix} \quad (\text{S31})$$

in the sublattice eigenbasis, with $\eta_j = (-1)^{j+1}$ and $E_{+,j} = \sqrt{h_0 + \eta_j h}$ (S28).

Writing out the two branches explicitly, the outer pair of bands $E_{\pm,1}$ ($j = 1, \eta_1 = +1$) reads

$$|\Psi_{\epsilon,1}\rangle = \frac{1}{\sqrt{2} \sqrt{2h(h + h_z)}} \begin{pmatrix} -\frac{2J}{E_{+,1}} [c_x(h + h_z) + c_y(h_x + ih_y)] \\ -\frac{2J}{E_{+,1}} [-s_y(h + h_z) + is_x(h_x + ih_y)] \\ \epsilon(h + h_z) \\ \epsilon(h_x + ih_y) \end{pmatrix}, \quad (\text{S32})$$

whose lower member $|\Psi_{-,1}\rangle$ ($\epsilon = -1$) is the ground state that is typically populated in cold-atom experiments realizing the quarter-flux Harper-Hofstadter model [103–105], and whose upper member $|\Psi_{+,1}\rangle$ ($\epsilon = +1$) is its chiral partner, obtained by flipping the sign of the two lower (sublattice-odd) components. The inner pair of bands $E_{\pm,2}$ ($j = 2, \eta_2 = -1$) reads

$$|\Psi_{\epsilon,2}\rangle = \frac{1}{\sqrt{2} \sqrt{2h(h - h_z)}} \begin{pmatrix} -\frac{2J}{E_{+,2}} [c_x(h - h_z) - c_y(h_x + ih_y)] \\ -\frac{2J}{E_{+,2}} [-s_y(h - h_z) - is_x(h_x + ih_y)] \\ \epsilon(h - h_z) \\ -\epsilon(h_x + ih_y) \end{pmatrix}. \quad (\text{S33})$$

Together with the formula (S31), Eqs. (S32) and (S33) provide the exact analytical expressions for all four eigenstates of the quarter-flux Harper-Hofstadter model in the sublattice eigenbasis. The corresponding states in the original site basis of $H(\mathbf{k})$ (2) are simply recovered by the fixed permutation $|\Psi\rangle = Q|\Psi_{\epsilon,j}\rangle$ (S21), which just swaps the components. We keep all expressions in the sublattice ordering in the following.

For the ground state $|\Psi_{-,1}\rangle$ (S32), inserting the explicit Bloch-vector components $h_x = 4J^2 c_x c_y$, $h_y = 4J^2 s_x s_y$, $h_z = 2J^2(c_{2x} - c_{2y})$ and $h = J^2 \sqrt{2Z}$ (S24) and simplifying the trigonometric factors yields the fully closed form

expression

$$|\Psi_{-,1}\rangle = \frac{1}{\sqrt{2}\sqrt{2\sqrt{2Z}[\sqrt{2Z}+2(c_{2x}-c_{2y})]}} \begin{pmatrix} -\frac{2}{\sqrt{4+\sqrt{2Z}}} [c_x(\sqrt{2Z}+4c_x^2)+2is_xs_{2y}] \\ -\frac{2}{\sqrt{4+\sqrt{2Z}}} [-s_y(\sqrt{2Z}+4s_y^2)+2ic_ys_{2x}] \\ -[\sqrt{2Z}+2(c_{2x}-c_{2y})] \\ -4(c_xc_y+is_xs_y) \end{pmatrix}, \quad (\text{S34})$$

with $Z = 6 + c_{4x} + c_{4y}$, where we used $h + h_z = J^2[\sqrt{2Z} + 2(c_{2x} - c_{2y})]$ and $E_{+,1} = J\sqrt{4 + \sqrt{2Z}}$. This is the exact analytical ground state of the quarter-flux Harper-Hofstadter model for the symmetric $2a \times 2a$ MUC (ii) [Fig. S1(b)] and solves $H_S(\mathbf{k})|\Psi_{-,1}\rangle = E_{-,1}(\mathbf{k})|\Psi_{-,1}\rangle$ (S21). The other three eigenstates follow analogously from Eqs. (S32) and (S33).

Finally, let us briefly comment on the degeneracy of the inner two bands $E_{\pm,2}$, which form a single superband that touches at zero energy wherever $\sigma_2 = E_{+,2} = \sqrt{h_0 - \hbar}$ vanishes, i.e. $h = h_0$, or equivalently $Z = 8$ ($c_{4x} = c_{4y} = 1$), at the high-symmetry points seen as band touchings in Fig. 1. At these Dirac points the construction $|u_2\rangle = D_{\mathbf{k}}|v_2\rangle/\sigma_2$ becomes ill-defined and the two states $|\Psi_{\pm,2}\rangle$ (S33) become degenerate, so that only the two-dimensional subspace they span is well-defined and any $U(2)$ rotation within it is an equally valid eigenbasis. Away from these points all four eigenstates (S31) are smooth, up to the coordinate (gauge) singularity of the north-pole parametrization at $h + \eta_j h_z \rightarrow 0$, which is present for all two-band systems anyway [165].

S5. QUANTUM GEOMETRY OF SUBLATTICE SYMMETRIC SYSTEMS.

The quantum geometric tensor of a state $|\Psi\rangle$ is defined as [2]

$$Q_{\mu\nu} = \langle \partial_\mu \Psi | (1 - |\Psi\rangle\langle\Psi|) | \partial_\nu \Psi \rangle = \langle \partial_\mu \Psi | \partial_\nu \Psi \rangle - A_\mu A_\nu = g_{\mu\nu} - \frac{i}{2} \Omega_{\mu\nu}, \quad (\text{S35})$$

where $A_\mu = i\langle\Psi|\partial_\mu\Psi\rangle$ is the (Abelian) Berry connection, and $g_{\mu\nu} = \text{Re}[Q_{\mu\nu}] = g_{\nu\mu}$ and $\Omega_{\mu\nu} = -2\text{Im}[Q_{\mu\nu}] = -\Omega_{\nu\mu}$ are the quantum metric and the Berry curvature, respectively. It is of course implicitly assumed that the state $|\Psi\rangle \equiv |\Psi(\boldsymbol{\lambda})\rangle$ depends on a set of parameters $\boldsymbol{\lambda} = (\lambda^1, \lambda^2, \dots, \lambda^N)$, and that $\partial_\mu \equiv \partial/\partial\lambda^\mu$, with $\mu \in \{1, 2, \dots, N\}$. For a system with an N -dimensional parameter space the quantum geometric tensor, being hermitian, has N^2 independent components.

We now derive the sublattice decomposition of the QGT quoted in the main text (6). Recall from the main text that the eigenstates of any sublattice symmetric H take the form (S13), with $|u_j\rangle \in \mathcal{H}_+$ and

$|v_j\rangle \in \mathcal{H}_-$ the left and right singular vectors of D sharing the singular value σ_j , and $\epsilon = \pm 1$ labeling the chiral energy pair $E_{\pm,j} = \pm\sigma_j$. Inserting (S13) into the definition (S35) and using $A_\mu = \frac{1}{2}(A_\mu^{(u_j)} + A_\mu^{(v_j)})$ together with $\langle \partial_\mu \Psi_{\epsilon,j} | \partial_\nu \Psi_{\epsilon,j} \rangle = \frac{1}{2}(\langle \partial_\mu u_j | \partial_\nu u_j \rangle + \langle \partial_\mu v_j | \partial_\nu v_j \rangle)$ yields

$$Q_{\mu\nu}^j = \frac{1}{2} \left(Q_{\mu\nu}^{(u_j)} + Q_{\mu\nu}^{(v_j)} \right) + \frac{1}{4} \Delta A_\mu^j \Delta A_\nu^j, \quad (\text{S36})$$

where $Q_{\mu\nu}^{(u_j)} = \langle \partial_\mu u_j | \partial_\nu u_j \rangle - A_\mu^{(u_j)} A_\nu^{(u_j)}$ is the QGT (S35) and $A_\mu^{(u_j)} = i\langle u_j | \partial_\mu u_j \rangle$ the Berry connection of $|u_j\rangle$ (similarly for v_j), and where

$$\Delta A_\mu^j \equiv A_\mu^{(u_j)} - A_\mu^{(v_j)} \quad (\text{S37})$$

is the sector connection difference. As stressed in the main text, ΔA_μ^j is gauge invariant only because the SVD relation $|u_j\rangle = D|v_j\rangle/\sigma_j$ ties the two sectors. Under $|v_j\rangle \rightarrow e^{i\chi}|v_j\rangle$ it forces $|u_j\rangle \rightarrow e^{i\chi}|u_j\rangle$, so both connections shift by $\partial_\mu \chi$ such that the difference is unchanged. Splitting (S36) into real and imaginary parts, and using that $\Delta A_\mu^j \Delta A_\nu^j$ is real and symmetric, gives the metric and curvature decomposition (7) of the main text, which we restate here for convenience,

$$g_{\mu\nu}^j = \frac{1}{2} \left(g_{\mu\nu}^{(u_j)} + g_{\mu\nu}^{(v_j)} \right) + \frac{1}{4} \Delta A_\mu^j \Delta A_\nu^j, \\ \Omega_{\mu\nu}^j = \frac{1}{2} \left(\Omega_{\mu\nu}^{(u_j)} + \Omega_{\mu\nu}^{(v_j)} \right). \quad (\text{S38})$$

Since $Q_{\mu\nu}^j$ is independent of ϵ , the QGT is always the same for the two chiral partners $|\Psi_{+,j}\rangle$ and $|\Psi_{-,j}\rangle$ as long as the energy pair is non-degenerate. The sector form (S36), in which the connection difference ΔA_μ^j enters the metric holds for arbitrary sublattice dimensions $d_\pm$ and, as shown below, extends to the non-Abelian QGT. This highlights the role of virtual-interband quantum metric effects [154] in sublattice-symmetric multi-band systems.

The sector contributions follow from the $M_{\mathbf{k}} = D_{\mathbf{k}}^\dagger D_{\mathbf{k}}$ and $\tilde{M}_{\mathbf{k}} = D_{\mathbf{k}} D_{\mathbf{k}}^\dagger$ machinery of Sec. S3, that is, $|v_j\rangle$ and $|u_j\rangle$ are also the eigenvectors of the two-level blocks M and $\tilde{M}$ (S27), with Bloch vectors $\mathbf{h}$ and $\tilde{\mathbf{h}}$ (S24). This allows us to utilize the established machinery of two-band quantum geometry [155, 166]. To this end, note that the quantum metric $g_{\mu\nu}$ and Berry curvature $\Omega_{\mu\nu}$ (S35)

of a generic two-level system are fully determined by a (normalized) Bloch vector $\hat{\mathbf{h}} = \mathbf{h}/h$ as those in (S24) and given by

$$\begin{aligned} g_{\mu\nu} &= \frac{1}{4} \left(\partial_\mu \hat{\mathbf{h}} \cdot \partial_\nu \hat{\mathbf{h}} \right) \\ &= \frac{1}{4h^4} \left[h^2 \partial_\mu \mathbf{h} \cdot \partial_\nu \mathbf{h} - (\mathbf{h} \cdot \partial_\mu \mathbf{h})(\mathbf{h} \cdot \partial_\nu \mathbf{h}) \right], \quad (\text{S39}) \\ \Omega_{\mu\nu} &= \mp \frac{1}{2} \hat{\mathbf{h}} \cdot (\partial_\mu \hat{\mathbf{h}} \times \partial_\nu \hat{\mathbf{h}}) = \mp \frac{1}{2h^3} \mathbf{h} \cdot (\partial_\mu \mathbf{h} \times \partial_\nu \mathbf{h}). \quad (\text{S40}) \end{aligned}$$

Note that for the two eigenstates of a two-level system pointing to $\pm \hat{\mathbf{h}}$ on the Bloch sphere, the quantum metric (S39) is the same, while the Berry curvature switches sign, as indicated by the $\mp$ in (S40). In our two-dimensional parameter space $\mathbf{k} = (k_x, k_y)$ the indices μ and ν take values in $\{x, y\}$, e.g., $\partial_x \equiv \partial/\partial k_x$, such that we have single independent component of the Berry curvature $\Omega_{xy} \equiv \Omega$ and three independent components of the quantum metric g_{xx} , g_{yy} , and g_{xy} . As we focus here first on the ground state $|\Psi_{-,1}\rangle$ (S32) (and its chirally symmetric partner, the state $|\Psi_{+,1}\rangle$ corresponding to the highest energy band), the relevant eigenvectors of M and $\tilde{M}$ corresponding to the eigenvalue $\lambda_1 = (E_{+,1})^2 = h_0 + h$ are $|v_1\rangle$ and $|u_1\rangle$ pointing to $+\hat{\mathbf{h}}$ and $+\hat{\tilde{\mathbf{h}}}$ respectively.

We proceed by first computing the derivatives $\partial_\mu \mathbf{h}$ and $\partial_\mu \tilde{\mathbf{h}}$ of the Bloch vectors (S24) which are given by,

$$\begin{aligned} \partial_x \mathbf{h} &= 4J^2 a \begin{pmatrix} -s_x c_y \\ c_x s_y \\ -s_{2x} \end{pmatrix}, \quad \partial_y \mathbf{h} = 4J^2 a \begin{pmatrix} -c_x s_y \\ s_x c_y \\ s_{2y} \end{pmatrix}, \\ \partial_x \tilde{\mathbf{h}} &= 4J^2 a \begin{pmatrix} s_x s_y \\ c_x c_y \\ -s_{2x} \end{pmatrix}, \quad \partial_y \tilde{\mathbf{h}} = -4J^2 a \begin{pmatrix} c_x c_y \\ s_x s_y \\ s_{2y} \end{pmatrix}. \quad (\text{S41}) \end{aligned}$$

Let us now first focus on the single Berry curvature Ω_{k_x, k_y} component. Using (S41) we first find that

$$\begin{aligned} \mathbf{h} \cdot (\partial_x \mathbf{h} \times \partial_y \mathbf{h}) &= 16J^6 a^2 [4 - (c_{2x} + c_{2y})^2], \\ \tilde{\mathbf{h}} \cdot (\partial_x \tilde{\mathbf{h}} \times \partial_y \tilde{\mathbf{h}}) &= 16J^6 a^2 [4 - (c_{2x} - c_{2y})^2]. \quad (\text{S42}) \end{aligned}$$

Thus, the Berry curvature contributions from the two sectors, $\Omega_{xy}^{(v_1)}$ and $\Omega_{xy}^{(u_1)}$, are given by

$$\Omega_{xy}^{(v_1/u_1)} = -\frac{2\sqrt{2}a^2[4 - (c_{2x} \pm c_{2y})^2]}{Z^{3/2}}, \quad (\text{S43})$$

where we have used $h = \tilde{h} = \sqrt{2}J^2\sqrt{Z}$ from above (S24). Here, the exponent (v_1/u_1) shall indicate the upper/lower $\pm$ sign in the above expression. From (S40) it directly follows that the corresponding Berry curvature contributions for the other sector are given by $\Omega_{xy}^{(v_2/u_2)} = -\Omega_{xy}^{(v_1/u_1)}$, which will become relevant later for the non-Abelian QGT of the remaining middle two energy bands $E_{\pm,2}$. Here, for the lowest band $E_{-,1}$, we finally arrive at the exact analytical for the full Berry curvature

$$\Omega_{xy}^1 = (\Omega_{xy}^{(u_1)} + \Omega_{xy}^{(v_1)})/2 \quad (\text{S38})$$

$$\begin{aligned} \Omega_{xy}^1 &= -\frac{\sqrt{2}a^2[12 - Z]}{Z^{3/2}} \\ &= -\sqrt{2}a^2 \frac{6 - \cos(4k_x a) - \cos(4k_y a)}{[6 + \cos(4k_x a) + \cos(4k_y a)]^{3/2}} \quad (\text{S44}) \end{aligned}$$

where we have used that $8 - 2(c_{2x}^2 + c_{2y}^2) = 12 - Z$ with $Z = 6 + c_{4x} + c_{4y}$ from above (S24). Eq. (S44) is the Berry curvature Ω_{xy}^1 for both $|\Psi_{\pm,1}\rangle$ (S32), i.e., the lowest and highest energy band of the quarter-flux Harper-Hofstadter model, which is shown in Fig. S2(d) and (h) as a function of $\mathbf{k}$ over the FBZ and along a high-symmetry path in quasimomentum space, respectively. Moreover, note that since $Z \in [4, 8]$ for all $\mathbf{k}$, we have $12 - Z > 0$ and hence $\Omega_{xy}^1 < 0$, $\forall \mathbf{k}$, we find that the Berry curvature does not switch sign over $\mathbf{k}$ -space and is uniformly negative, consistent with the Chern number $C = -1$ of the lowest and highest energy band. Our result (S44) is consistent with Ref. [111], where it was found, perturbatively, that the Berry curvature of the N -band Harper-Hofstadter model, i.e., for flux $\Phi = 2\pi/N$ (here $N = 4$ bands) admits a $\cos(Nk_x a) + \cos(Nk_y a)$ -dependence.

Let us now turn our attention to the quantum metric $g_{\mu\nu}$ components. From (S41) we first find the following inner products between the Bloch vectors themselves and their derivatives:

$$\begin{aligned} \partial_x \mathbf{h} \cdot \partial_y \mathbf{h} &= -8J^4 a^2 s_{2x} s_{2y}, \\ \partial_x \tilde{\mathbf{h}} \cdot \partial_y \tilde{\mathbf{h}} &= 8J^4 a^2 s_{2x} s_{2y}, \\ |\partial_\mu \mathbf{h}|^2 &= 8J^4 a^2 (2 - c_{2x} c_{2y} - c_{4\mu}), \\ |\partial_\mu \tilde{\mathbf{h}}|^2 &= 8J^4 a^2 (2 + c_{2x} c_{2y} - c_{4\mu}), \\ \frac{1}{2} \partial_\mu h^2 &= \mathbf{h} \cdot \partial_\mu \mathbf{h} = -4J^4 a s_{4\mu} = \tilde{\mathbf{h}} \cdot \partial_\mu \tilde{\mathbf{h}}, \quad (\text{S45}) \end{aligned}$$

where in the last expression we have used that $h^2 = \tilde{h}^2 = 2J^4 Z$ (S24). Using these results we arrive at the individual contributions of the two sectors to the quantum metric $g_{\mu\nu}^{(v_1)}$ and $g_{\mu\nu}^{(u_1)}$ (S39) given by

$$g_{\mu\nu}^{(v_1/u_1)} = \frac{a^2}{Z^2} \left[Z \mathcal{N}_{\mu\nu}^{(v_1/u_1)} - s_{4\mu} s_{4\nu} \right], \quad (\text{S46})$$

where we introduced the shorthand notation $\mathcal{N}_{\mu\nu}^{(v_1/u_1)}$ defined by

$$\begin{aligned} \mathcal{N}_{\mu\mu}^{(v_1)} &= 2 - c_{2x} c_{2y} - c_{4\mu}, & \mathcal{N}_{xy}^{(v_1)} &= -s_{2x} s_{2y}, \\ \mathcal{N}_{\mu\mu}^{(u_1)} &= 2 + c_{2x} c_{2y} - c_{4\mu}, & \mathcal{N}_{xy}^{(u_1)} &= s_{2x} s_{2y}, \quad (\text{S47}) \end{aligned}$$

with $\mu \in \{x, y\}$ and no summation over repeated indices. From (S39) it follows that $g_{\mu\nu}^{(v_1)} = g_{\mu\nu}^{(v_2)}$ and $g_{\mu\nu}^{(u_1)} = g_{\mu\nu}^{(u_2)}$ as well, which will become relevant later when we compute QGT of the degenerate middle two energy bands.

Now, what remains to be computed for the full quantum metric $g_{\mu\nu}$ (S38) is the contribution from the connection difference $\Delta A_\mu^1 = A_\mu^{(u_1)} - A_\mu^{(v_1)}$, which we will

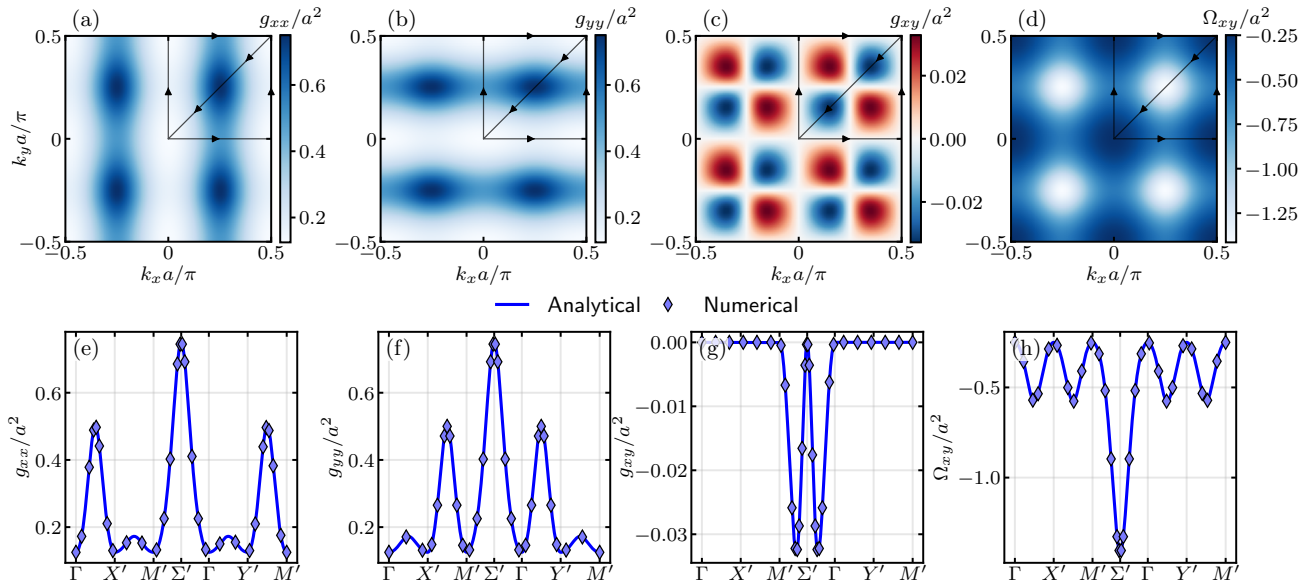


FIG. S2. Quantum metric components and Berry curvature of the lowest (and highest) energy band of the quarter-flux Harper-Hofstadter model. (a)-(d) Analytically obtained quantum metric components g_{xx}^1 (S52), g_{yy}^1 (S53), g_{xy}^1 (S54), and Berry curvature Ω_{xy}^1 (S44) as function of $\mathbf{k}$ over the FBZ, for the lowest and highest energy band $E_{\pm,1}$, corresponding to the eigenstate $|\Psi_{\pm,1}\rangle$ (S32). (e)-(h) Same as (a)-(d) but along the high-symmetry path $\Gamma-X'-M'-\Sigma'-\Gamma-Y'-M'$ (as indicated via solid lines with arrows in (a)-(d), see also Fig. 1 in the main text) where solid lines (diamonds) depict the analytical (numerical) results.

derive in the next step. To this end, we first express two eigenstates $|v_j\rangle$ of M (Eq. (S29), $j \in \{1, 2\}$) in terms of their Bloch vector $\mathbf{h}$ components as [165]

$$|v_j\rangle = \frac{1}{\sqrt{2h(h + \eta_j h_z)}} \begin{pmatrix} h + \eta_j h_z \\ \eta_j(h_x + i h_y) \end{pmatrix}, \quad (\text{S48})$$

where again $\eta_j = (-1)^{j+1}$. As discussed above, $|u_j\rangle$ must be expressed in the same gauge via the SVD relation $|u_j\rangle = D|v_j\rangle/\sigma_j$. Using this relation and the eigenvalue equation $D^\dagger D|v_j\rangle = M|v_j\rangle = \sigma_j^2|v_j\rangle$ (S27), the connection differences ΔA_μ^j (S37) take the form of

$$\begin{aligned} \Delta A_\mu^j &= \frac{i}{\sigma_j} \langle v_j | D^\dagger \partial_\mu \left(\frac{1}{\sigma_j} D |v_j\rangle \right) - i \langle v_j | \partial_\mu v_j \rangle \\ &= i \left[\frac{1}{\sigma_j^2} \langle v_j | D^\dagger \partial_\mu D |v_j\rangle - \frac{\partial_\mu \sigma_j}{\sigma_j} \right] \end{aligned} \quad (\text{S49})$$

where we find that the Berry connection $A_\mu^{(v_j)}$ cancels out exactly. Using that $\partial_\mu \sigma_j / \sigma_j = (\eta_j \partial_\mu \mathbf{h} \cdot \mathbf{h}) / 2h\sigma_j^2$ and $D^\dagger \partial_x D = -4J^2 \begin{pmatrix} s_{2x}/2 & i c_x s_y \\ c_y s_x & -s_{2x}/2 \end{pmatrix}$ and $D^\dagger \partial_y D = 4J^2 \begin{pmatrix} s_{2y}/2 & -c_x s_y \\ i s_x c_y & -s_{2y}/2 \end{pmatrix}$, we first arrive at the intermediate re-

sults of

$$\begin{aligned} \frac{\langle v_j | D^\dagger \partial_x D |v_j\rangle}{\sigma_j^2} &= \frac{-2\eta_j J^4 a}{h\sigma_j^2} (s_{4x} + 2i s_{2y} c_{2x}), \\ \frac{\langle v_j | D^\dagger \partial_y D |v_j\rangle}{\sigma_j^2} &= \frac{2\eta_j J^4 a}{h\sigma_j^2} (-s_{4y} + 2i s_{2x} c_{2y}), \\ \frac{\partial_\mu \sigma_j}{\sigma_j} &= -2\eta_j J^4 a \frac{s_{4\mu}}{h\sigma_j^2}. \end{aligned} \quad (\text{S50})$$

Thus, the exact expressions for the Berry connection differences ΔA_μ^j (S49) become

$$\Delta A_x^j = \frac{2\eta_j a}{\mathcal{G}_j} s_{2y} c_{2x}, \quad \Delta A_y^j = -\frac{2\eta_j a}{\mathcal{G}_j} s_{2x} c_{2y}, \quad (\text{S51})$$

with $\mathcal{G}_j = 2\sqrt{2Z} + \eta_j Z$ for brevity. Finally, using these results and (S46), we arrive at exact analytical expressions for the quantum metric components $g_{\mu\nu}^1$ (S38)

$$g_{xx}^1/a^2 = \frac{Z(2 - c_{4x}) - s_{4x}^2}{Z^2} + \frac{s_{2y}^2 c_{2x}^2}{(\mathcal{G}_1)^2}, \quad (\text{S52})$$

$$g_{yy}^1/a^2 = \frac{Z(2 - c_{4y}) - s_{4y}^2}{Z^2} + \frac{s_{2x}^2 c_{2y}^2}{(\mathcal{G}_1)^2}, \quad (\text{S53})$$

$$g_{xy}^1/a^2 = -\frac{s_{4x} s_{4y}}{Z^2} - \frac{s_{2x} s_{2y} c_{2x} c_{2y}}{(\mathcal{G}_1)^2} \quad (\text{S54})$$

Eqs. (S52) to (S54) are the exact analytical expressions for the quantum metric components $g_{\mu\nu}^1$ of the

lowest and highest energy band of the quarter-flux Harper-Hofstadter model corresponding to the eigenstates $|\Psi_{\pm,1}\rangle$ (S32). These components are depicted in Fig. S2(a)-(c) and (e)-(g) as function of $\mathbf{k}$ over the FBZ and along a high-symmetry path in quasimomentum space, respectively. Summing the diagonal components gives the metric trace in closed form,

$$\frac{\text{tr } g^1(\mathbf{k})}{a^2} = \frac{Z(10-Z) - 2 + c_{4x}^2 + c_{4y}^2}{Z^2} + \frac{s_{2y}^2 c_{2x}^2 + s_{2x}^2 c_{2y}^2}{(2\sqrt{2}Z + Z)^2}, \quad (\text{S55})$$

where we used $c_{4x} + c_{4y} = Z - 6$, and where the second term is the connection-difference contribution $\frac{1}{4}((\Delta A_x^1)^2 + (\Delta A_y^1)^2)$ (S51).

The determinant $\det g^1$ follows in closed form as well. Writing the decomposition (S38) as $g_{\mu\nu}^1 = \bar{g}_{\mu\nu}^1 + \frac{1}{4}\Delta A_\mu^1 \Delta A_\nu^1$ with the sector average $\bar{g}_{\mu\nu}^1 \equiv \frac{1}{2}(g_{\mu\nu}^{(u_1)} + g_{\mu\nu}^{(v_1)})$ (S46), the connection-difference term is an outer product, so its own determinant vanishes. Thus, we have

$$\det g^1 = \det \bar{g}^1 + \frac{1}{4}[\bar{g}_{yy}^1 (\Delta A_x^1)^2 - 2\bar{g}_{xy}^1 \Delta A_x^1 \Delta A_y^1 + \bar{g}_{xx}^1 (\Delta A_y^1)^2], \quad (\text{S56})$$

with ΔA_μ^1 given by (S51) and $\bar{g}_{\mu\nu}^1$ given by (S46), and

$$\frac{\det \bar{g}^1}{a^4} = \frac{(2 - c_{4x})(2 - c_{4y})}{Z^2} - \frac{(2 - c_{4y})s_{4x}^2 + (2 - c_{4x})s_{4y}^2}{Z^3}. \quad (\text{S57})$$

Both the determinant $\det g^1(\mathbf{k})$ and the trace $\text{tr } g^1(\mathbf{k})$ (S55) are depicted in Fig. S4(a)-(b) and (e)-(f), respectively.

A. Non-Abelian Quantum Geometry of sublattice symmetric systems

The middle two bands (one super band) of the quarter-flux Harper-Hofstadter model exhibit band touching points (Dirac points) at zero energy at certain high-symmetry points in the Brillouin zone [see Fig. 1 in the main text], and are hence degenerate and not gapped for all $\mathbf{k}$. Thus, the quantum geometry of these two bands cannot be described by the QGT of an isolated single band (S35), but rather requires the notion of quantum geometry for a degenerate subspace. This is captured by the so-called non-Abelian quantum geometric

tensor [12, 167]

$$\begin{aligned} \mathcal{Q}_{\mu\nu}^{ab} &= \langle \partial_\mu \Psi_a | (1 - P) | \partial_\nu \Psi_b \rangle \\ &= \langle \partial_\mu \Psi_a | \partial_\nu \Psi_b \rangle - \sum_c A_\mu^{ac} A_\nu^{cb} \\ &= G_{\mu\nu}^{ab} - \frac{i}{2} F_{\mu\nu}^{ab}, \end{aligned} \quad (\text{S58})$$

where a, b, c are indices labeling the states within the degenerate subspace, and P the projector onto the degenerate subspace, $A_\mu^{ab} = i\langle \Psi_a | \partial_\mu \Psi_b \rangle = (A_\mu^{ba})^*$ the non-Abelian (or interband) Berry connection. For fixed parameter-indices μ, ν , the non-Abelian QGT $\mathcal{Q}_{\mu\nu}$ can be understood as $M \times M$ matrix where M labels the dimension of the degenerate subspace which further satisfies $\mathcal{Q}_{\mu\nu}^\dagger = \mathcal{Q}_{\nu\mu}$ (or $\mathcal{Q}_{\mu\nu}^{ab} = (\mathcal{Q}_{\nu\mu}^{ba})^*$). It can be decomposed into a Hermitian symmetric part, $G_{\mu\nu} = (\mathcal{Q}_{\mu\nu} + \mathcal{Q}_{\mu\nu}^\dagger)/2 = G_{\nu\mu} = G_{\mu\nu}^\dagger$, the non-Abelian quantum metric, and a Hermitian antisymmetric part, $F_{\mu\nu} = i(\mathcal{Q}_{\mu\nu} - \mathcal{Q}_{\mu\nu}^\dagger) = -F_{\nu\mu} = F_{\mu\nu}^\dagger$, the non-Abelian Berry curvature. The latter can be further written as the field strength of the non-Abelian Berry connection A_μ [168] via

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - i[A_\mu, A_\nu], \quad (\text{S59})$$

which is to be understood as a $M \times M$ matrix-valued differential 2-form. Note that under a local $U(N)$ change of basis inside the degenerate subspace, $|\Psi_a\rangle \rightarrow \sum_b U_{ab} |\Psi_b\rangle$, the non-Abelian QGT (S58) transforms covariantly, $\mathcal{Q}_{\mu\nu} \rightarrow U \mathcal{Q}_{\mu\nu} U^\dagger$ so that only similarity invariants of the non-Abelian QGT, such as the trace $\text{tr}[\mathcal{Q}_{\mu\nu}]$ and determinant $\det[\mathcal{Q}_{\mu\nu}]$, are truly gauge invariant and can be used to characterize the quantum geometry of the degenerate subspace.

In the following, we will first show how the non-Abelian quantum geometric tensor decomposes in the present case of a sublattice symmetric system described by eigenstates of the form of (S13), which we will then apply to obtain the full non-Abelian QGT for the degenerate subspace of the middle two energy bands of the quarter-flux Harper-Hofstadter model.

For a sublattice symmetric system described by eigenstates of the form of (S13), using $a \equiv (\epsilon', j')$ and $b \equiv (\epsilon, j)$, we first have that the non-Abelian Berry connection $A_\mu^{ab} = A_\mu^{(\epsilon', j'), (\epsilon, j)}$ becomes

$$A_\mu^{(\epsilon', j'), (\epsilon, j)} = \frac{1}{2} \left[A_\mu^{(u_{j'}, u_j)} + \epsilon' \epsilon A_\mu^{(v_{j'}, v_j)} \right], \quad (\text{S60})$$

where $A_\mu^{(u_{j'}, u_j)} = i\langle u_{j'} | \partial_\mu u_j \rangle$ and similarly for $A_\mu^{(v_{j'}, v_j)}$, are interband Berry connections within the $\mathcal{H}_+$ and $\mathcal{H}_-$ sectors, respectively. Moreover, using that $\langle \partial_\mu \Psi_{(\epsilon', j')} | \partial_\nu \Psi_{(\epsilon, j)} \rangle = \frac{1}{2} [\langle \partial_\mu u_{j'} | \partial_\nu u_j \rangle + \epsilon' \epsilon \langle \partial_\mu v_{j'} | \partial_\nu v_j \rangle]$, we can see that the full non-Abelian QGT (S58) in a sublattice symmetric decomposes into contributions from the two sectors as

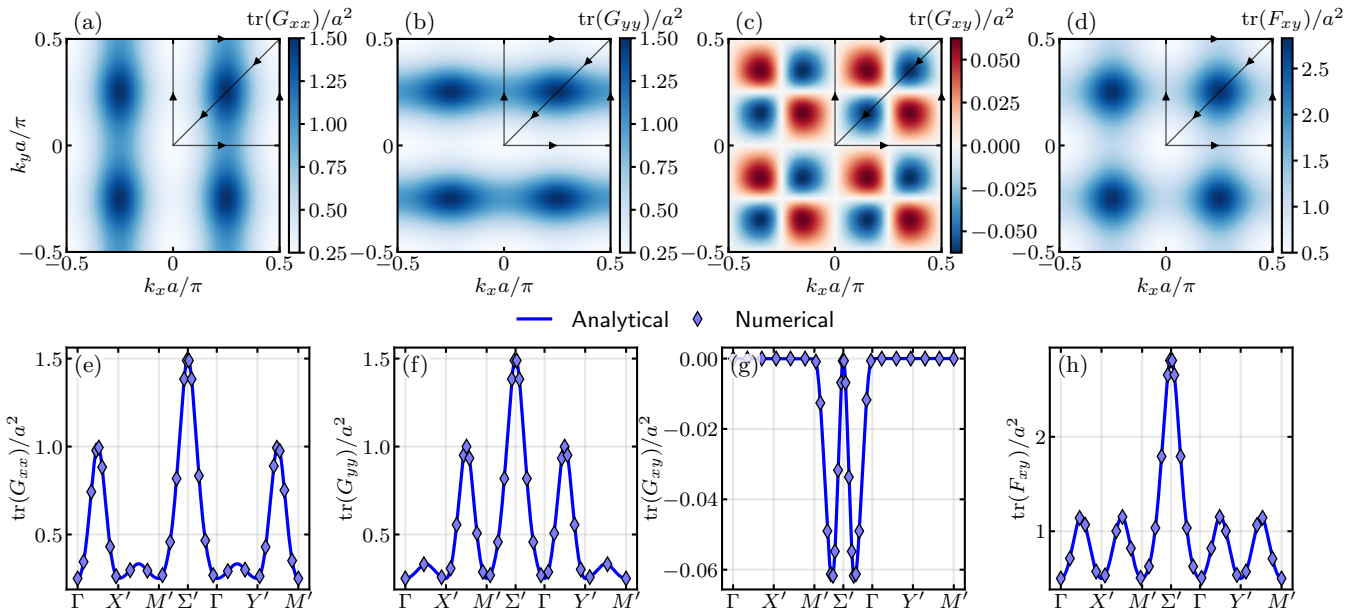


FIG. S3. Non-Abelian quantum metric and Berry curvature of the degenerate subspace of the middle two energy bands $E_{\pm,2}$ of the quarter-flux Harper-Hofstadter model. (a)-(d) Analytically obtained non-Abelian quantum metric components $\text{tr}(G_{xx})$, $\text{tr}(G_{yy})$, $\text{tr}(G_{xy})$ (S68), and Berry curvature $\text{tr}(F_{xy})$ (S71) as function of $\mathbf{k}$ over the FBZ, for the degenerate subspace of the middle two energy bands $E_{\pm,2}$. (e)-(h) Same as (a)-(d) but along the high-symmetry path $\Gamma-X'-M'-\Sigma'-\Gamma-Y'-M'$ (as indicated via solid lines with arrows in (a)-(d), see also Fig. 1 in the main text) where solid lines (diamonds) depict the analytical (numerical) results.

$$\begin{aligned}
\mathcal{Q}_{\mu\nu}^{(\epsilon',j'),(\epsilon,j)} &= \frac{1}{2} [\langle \partial_\mu u_{j'} | \partial_\nu u_j \rangle + \epsilon' \epsilon \langle \partial_\mu v_{j'} | \partial_\nu v_j \rangle] - \frac{1}{4} \sum_{\epsilon'',j''} \left[A_\mu^{(u_{j'},u_{j''})} + \epsilon' \epsilon'' A_\mu^{(v_{j'},v_{j''})} \right] \left[A_\nu^{(u_{j'',u_j})} + \epsilon'' \epsilon A_\nu^{(v_{j'',v_j})} \right] \\
&= \frac{1}{2} \left[\langle \partial_\mu u_{j'} | \partial_\nu u_j \rangle - \sum_{j''} A_\mu^{(u_{j'},u_{j''})} A_\nu^{(u_{j'',u_j})} \right] + \frac{\epsilon' \epsilon}{2} \left[\langle \partial_\mu v_{j'} | \partial_\nu v_j \rangle - \sum_{j''} A_\mu^{(v_{j'},v_{j''})} A_\nu^{(v_{j'',v_j})} \right] \\
&= \frac{1}{2} \left[\mathcal{Q}_{\mu\nu}^{(u_{j'},u_j)} + \epsilon' \epsilon \mathcal{Q}_{\mu\nu}^{(v_{j'},v_j)} \right].
\end{aligned} \tag{S61}$$

Here, we have used that $(\epsilon'')^2 = 1$ and $\mathcal{Q}_{\mu\nu}^{(u_{j'},u_j)} = \langle \partial_\mu u_{j'} | \partial_\nu u_j \rangle - \sum_{j''} A_\mu^{(u_{j'},u_{j''})} A_\nu^{(u_{j'',u_j})}$ and similarly for $\mathcal{Q}_{\mu\nu}^{(v_{j'},v_j)}$.

In the second line above (S61) we have used that

$$\begin{aligned}
&\sum_{\epsilon'',j''} \left[A_\mu^{(u_{j'},u_{j''})} + \epsilon' \epsilon'' A_\mu^{(v_{j'},v_{j''})} \right] \left[A_\nu^{(u_{j'',u_j})} + \epsilon'' \epsilon A_\nu^{(v_{j'',v_j})} \right] \\
&= \sum_{j''} \left[\sum_{\epsilon''=\pm} \left\{ \left(A_\mu^{(u_{j'},u_{j''})} A_\nu^{(u_{j'',u_j})} + \epsilon' \epsilon A_\mu^{(v_{j'},v_{j''})} A_\nu^{(v_{j'',v_j})} \right) + \epsilon'' \left(A_\mu^{(u_{j'},u_{j''})} A_\nu^{(v_{j'',v_j})} + \epsilon' \epsilon A_\mu^{(v_{j'},v_{j''})} A_\nu^{(u_{j'',u_j})} \right) \right\} \right] \\
&= 2 \sum_{j''} \left(A_\mu^{(u_{j'},u_{j''})} A_\nu^{(u_{j'',u_j})} + \epsilon' \epsilon A_\mu^{(v_{j'},v_{j''})} A_\nu^{(v_{j'',v_j})} \right)
\end{aligned} \tag{S62}$$

In the $(\epsilon, \epsilon') \in \{\pm, \pm\}$ basis, the non-Abelian QGT $\mathcal{Q}_{\mu\nu}^{(\epsilon',j'),(\epsilon,j)}$ (S61) takes the form of

$$\mathcal{Q}_{\mu\nu} = \begin{pmatrix} \mathcal{Q}_{\mu\nu}^{(+,j'),(+,j)} & \mathcal{Q}_{\mu\nu}^{(+,j'),(-,j)} \\ \mathcal{Q}_{\mu\nu}^{(-,j'),(+,j)} & \mathcal{Q}_{\mu\nu}^{(-,j'),(-,j)} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} \mathcal{Q}_{\mu\nu}^{(u_{j'},u_j)} + \mathcal{Q}_{\mu\nu}^{(v_{j'},v_j)} & \mathcal{Q}_{\mu\nu}^{(u_{j'},u_j)} - \mathcal{Q}_{\mu\nu}^{(v_{j'},v_j)} \\ \mathcal{Q}_{\mu\nu}^{(u_{j'},u_j)} - \mathcal{Q}_{\mu\nu}^{(v_{j'},v_j)} & \mathcal{Q}_{\mu\nu}^{(u_{j'},u_j)} + \mathcal{Q}_{\mu\nu}^{(v_{j'},v_j)} \end{pmatrix}. \tag{S63}$$

Let us now note how the diagonal parts $\mathcal{Q}_{\mu\nu}^{(\epsilon,j),(\epsilon,j)}$ reduce to the Abelian QGT (S35) for the case of an isolated

single band. Given the projector $P_j = |\Psi_{j,+}\rangle\langle\Psi_{j,+}| + |\Psi_{j,-}\rangle\langle\Psi_{j,-}|$ onto the two orthogonal and chirally symmetric pair of bands $E_{\pm,j}$ and using the relation $1 - |\Psi_{j,-}\rangle\langle\Psi_{j,-}| = (1 - P_j) + |\Psi_{j,+}\rangle\langle\Psi_{j,+}|$ we obtain that

$$\begin{aligned} \mathcal{Q}_{\mu\nu}^{(-,j),(-,j)} &= \langle\partial_\mu\Psi_{-,j}|\partial_\nu\Psi_{-,j}\rangle \\ &= Q_{\mu\nu}^j - \langle\partial_\mu\Psi_{-,j}|\Psi_{+,j}\rangle\langle\Psi_{+,j}|\partial_\nu\Psi_{-,j}\rangle \\ &= Q_{\mu\nu}^j - \frac{1}{4}\Delta A_\mu^j\Delta A_\nu^j, \end{aligned} \quad (\text{S64})$$

where in the second line we have used $Q_{\mu\nu}^j = \langle\partial_\mu\Psi_{-,j}|\partial_\nu\Psi_{-,j}\rangle$ (S35) and $\langle\partial_\mu\Psi_{-,j}|\Psi_{+,j}\rangle = i\Delta A_\mu^j/2$ (S37). Thus, with (S63) we indeed have

$$\begin{aligned} Q_{\mu\nu}^j &= \mathcal{Q}_{\mu\nu}^{(-,j),(-,j)} + \frac{1}{4}\Delta A_\mu^j\Delta A_\nu^j, \\ &= \frac{1}{2}[\mathcal{Q}_{\mu\nu}^{(u_j,u_j)} + \mathcal{Q}_{\mu\nu}^{(v_j,v_j)}] + \frac{1}{4}\Delta A_\mu^j\Delta A_\nu^j, \end{aligned} \quad (\text{S65})$$

matching our previous result in Eq. (S36) with $\mathcal{Q}_{\mu\nu}^{(u_j,u_j)} \equiv Q_{\mu\nu}^{(u_j)}$ (similarly for v_j).

For our specific case of degenerate subspace of the middle two energy bands, we have that projector reads $P = \sum_{\epsilon=\pm}|\Psi_{\epsilon,2}\rangle\langle\Psi_{\epsilon,2}| = |u_2\rangle\langle u_2| \oplus |v_2\rangle\langle v_2|$, such the non-Abelian QGT $\mathcal{Q}_{\mu\nu}^{(\epsilon',2),(\epsilon,2)}$ becomes 2×2 matrix-form for each pair of parameter indices μ, ν with matrix elements given by (S61) as

$$\begin{aligned} \mathcal{Q}_{\mu\nu} &= \begin{pmatrix} \mathcal{Q}_{\mu\nu}^{(+,2),(+,2)} & \mathcal{Q}_{\mu\nu}^{(+,2),(-,2)} \\ \mathcal{Q}_{\mu\nu}^{(-,2),(+,2)} & \mathcal{Q}_{\mu\nu}^{(-,2),(-,2)} \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} \mathcal{Q}_{\mu\nu}^{(u_2,u_2)} + \mathcal{Q}_{\mu\nu}^{(v_2,v_2)} & \mathcal{Q}_{\mu\nu}^{(u_2,u_2)} - \mathcal{Q}_{\mu\nu}^{(v_2,v_2)} \\ \mathcal{Q}_{\mu\nu}^{(u_2,u_2)} - \mathcal{Q}_{\mu\nu}^{(v_2,v_2)} & \mathcal{Q}_{\mu\nu}^{(u_2,u_2)} + \mathcal{Q}_{\mu\nu}^{(v_2,v_2)} \end{pmatrix} \\ &= \frac{1}{2}[(\mathcal{Q}_{\mu\nu}^{(u_2,u_2)} + \mathcal{Q}_{\mu\nu}^{(v_2,v_2)})\mathbb{1}_2 + (\mathcal{Q}_{\mu\nu}^{(u_2,u_2)} - \mathcal{Q}_{\mu\nu}^{(v_2,v_2)})\sigma_x]. \end{aligned} \quad (\text{S66})$$

Now, we proceed as before noting that $\mathcal{Q}_{\mu\nu}^{(u_2,u_2)} \equiv Q_{\mu\nu}^{(u_2)} = g_{\mu\nu}^{(u_2)} - \frac{i}{2}\Omega_{\mu\nu}^{(u_2)}$ (similarly for v_2) can be computed using two-level QGT (S39)-(S40) since $|u_2\rangle$ and $|v_2\rangle$, being the left and right singular vectors of D , are also eigenstates of 2-by-2 matrices M and $\tilde{M}$ (S27), respectively. The quantum metric components are readily obtained from noting that $g_{\mu\nu}^{(u_2)} = g_{\mu\nu}^{(u_1)}$ and $g_{\mu\nu}^{(v_2)} = g_{\mu\nu}^{(v_1)}$ (S46). Thus, using $\mathcal{Q}_{\mu\nu} = G_{\mu\nu} - \frac{i}{2}F_{\mu\nu}$ (S58), here we have, first for the quantum metric

$$G_{\mu\nu} = \frac{1}{2}[(g_{\mu\nu}^{(u_2)} + g_{\mu\nu}^{(v_2)})\mathbb{1}_2 + (g_{\mu\nu}^{(u_2)} - g_{\mu\nu}^{(v_2)})\sigma_x]. \quad (\text{S67})$$

The individual components are given by

$$\begin{aligned} G_{xx}/a^2 &= \frac{1}{Z^2}[(Z(2 - c_{4x}) - s_{4x}^2)\mathbb{1}_2 + (Zc_{2x}c_{2y})\sigma_x], \\ G_{yy}/a^2 &= \frac{1}{Z^2}[(Z(2 - c_{4y}) - s_{4y}^2)\mathbb{1}_2 + (Zc_{2x}c_{2y})\sigma_x], \\ G_{xy}/a^2 &= \frac{1}{Z^2}[(-s_{4x}s_{4y})\mathbb{1}_2 + (Zs_{2x}s_{2y})\sigma_x]. \end{aligned} \quad (\text{S68})$$

The total quantum metric of the degenerate subspace of the middle two energy bands is given by the trace of $G_{\mu\nu}$ over the internal degrees of freedom, $\text{tr}(G_{\mu\nu}) = g_{\mu\nu}^{(u_2)} + g_{\mu\nu}^{(v_2)}$, which is given by twice the components of (S68) proportional to the identity matrix,

$$\begin{aligned} \text{tr}(G_{xx})/a^2 &= \frac{2[Z(2 - c_{4x}) - s_{4x}^2]}{Z^2}, \\ \text{tr}(G_{yy})/a^2 &= \frac{2[Z(2 - c_{4y}) - s_{4y}^2]}{Z^2}, \\ \text{tr}(G_{xy})/a^2 &= \frac{-2s_{4x}s_{4y}}{Z^2}. \end{aligned} \quad (\text{S69})$$

These components are depicted in Fig. S3(a)-(c) and (e)-(g) as function of $\mathbf{k}$ over the FBZ and along a high-symmetry path in quasimomentum space, respectively. At this point, let us note that we observe the commonly overlooked property that the total quantum metric is non-additive [18, 154]. Namely in our case we have that $\text{tr}(G_{\mu\nu}) = g_{\mu\nu}^{(u_2)} + g_{\mu\nu}^{(v_2)} = g_{\mu\nu}^{(-,2)} + g_{\mu\nu}^{(+,2)} + \text{tr}(\partial_\mu P_{-,2}\partial_\nu P_{+,2}) \neq g_{\mu\nu}^{(-,2)} + g_{\mu\nu}^{(+,2)}$, where $g_{\mu\nu}^{(\pm,2)} \equiv g_{\mu\nu}^2$ (S38), $P_{\pm,2} = |\Psi_{\pm,2}\rangle\langle\Psi_{\pm,2}|$. Here, $\text{tr}(\partial_\mu P_{-,2}\partial_\nu P_{+,2}) = 2\text{Re}[\langle\Psi_{+,2}|\partial_\mu\Psi_{-,2}\rangle\langle\Psi_{-,2}|\partial_\nu\Psi_{+,2}\rangle] = -\Delta A_\mu^2\Delta A_\nu^2/2$ is the non-additive correction term involving the Berry connection differences (S37) (already appearing in the Abelian quantum metric of the individual bands (S38)) and which can be interpreted as virtual interband excitations between the two bands in the degenerate subspace [154]. This is to be contrasted with the trace of the non-Abelian Berry curvature, which is additive in the two constituents, as we will see in the following.

Now, finally for Berry curvature components we have that $\Omega_{\mu\nu}^{(u_2)} = -\Omega_{\mu\nu}^{(u_1)}$ and $\Omega_{\mu\nu}^{(v_2)} = -\Omega_{\mu\nu}^{(v_1)}$ (S43) such the non-Abelian Berry curvature components become

$$F_{\mu\nu} = \frac{1}{2}[(\Omega_{\mu\nu}^{(u_2)} + \Omega_{\mu\nu}^{(v_2)})\mathbb{1}_2 + (\Omega_{\mu\nu}^{(u_2)} - \Omega_{\mu\nu}^{(v_2)})\sigma_x]. \quad (\text{S70})$$

The individual components are given by

$$F_{xy}/a^2 = \frac{\sqrt{2}[12 - Z]}{Z^{3/2}}\mathbb{1}_2 + \frac{4\sqrt{2}}{Z^{3/2}}c_{2x}c_{2y}\sigma_x. \quad (\text{S71})$$

We find that the total Berry curvature of the degenerate subspace of the middle two energy bands, given by the trace $\text{tr}(F_{xy}) = \Omega_{xy}^{(u_2)} + \Omega_{xy}^{(v_2)} = -(\Omega_{xy}^{(u_1)} + \Omega_{xy}^{(v_1)})$, is given by

$$\begin{aligned} \text{tr}(F_{xy})/a^2 &= \frac{2\sqrt{2}[12 - Z]}{Z^{3/2}}, \\ &= 2\sqrt{2}\frac{6 - \cos(4k_x a) - \cos(4k_y a)}{[6 + \cos(4k_x a) + \cos(4k_y a)]^{3/2}}. \end{aligned} \quad (\text{S72})$$

This non-Abelian Berry curvature is depicted in Fig. S3(d) and (h) as function of $\mathbf{k}$ over the FBZ

and along a high-symmetry path in quasimomentum space, respectively. Note that this expression is both consistent with the additivity of the Berry curvature, i.e., $\text{tr}(F_{xy}) = \Omega_{xy}^{(u_2)} + \Omega_{xy}^{(v_2)} = \Omega_{xy}^{(-,2)} + \Omega_{xy}^{(+,2)}$ with $\Omega_{xy}^{(\pm,2)} \equiv \Omega_{xy}^2$ (S38) and with the conservation law of the total Berry curvature, i.e., $\text{tr}(F_{xy}) + \Omega_{xy}^1 + \Omega_{xy}^1 = \Omega_{xy}^{(u_2)} + \Omega_{xy}^{(v_2)} + 2 \cdot \frac{1}{2} (\Omega_{xy}^{(u_1)} + \Omega_{xy}^{(v_1)}) = -(\Omega_{xy}^{(u_1)} + \Omega_{xy}^{(v_1)}) + (\Omega_{xy}^{(u_1)} + \Omega_{xy}^{(v_1)}) = 0$ (S44), i.e., the total Berry curvature over all bands must vanish [17]. As $\text{tr}(F_{xy}) = -2\Omega_{xy}^1$ equals twice the negative value of the single-band Berry curvature $\Omega_{xy}^1 < 0$ (S44) of the lowest band, which has a Chern number of $C_1 = -1 = C_4$, we find that the total Chern number of the middle two bands to be $C_{2+3} = (2\pi)^{-1} \int_{\text{BZ}} d^2k \text{tr}(F_{xy}) = 2$, which is of course also consistent with conservation law that the sum over all Chern numbers must vanish, i.e., $C_1 + C_{2+3} + C_4 = 0$ [17].

S6. FCI STABILITY CONDITIONS

A number of criteria have recently been proposed to quantify the stability of a platform to host fractional Chern insulators (FCIs) or fractional quantum Hall (FQH) trial states. Interestingly, these criteria can be evaluated from the underlying single-particle wave functions where most of them rest on lowest-Landau-level (LLL) “mimicry” [91].

Equipped with the exact solutions and quantum geometry of the quarter-flux Harper-Hofstadter model, we can evaluate these criteria explicitly. We restrict ourselves to the quantum geometric momentum-space conditions, which have been originally formulated in Refs. [58, 79, 90] and further sharpened in Refs. [59, 60, 94, 97, 130, 132, 154, 173, 174]. We note that recently, complementary real-space criteria, in particular vortexability [91–93], have also been proposed, but are not the focus of this work. There are three notions which enter these criteria, which summarize below and go into more detail in the following.

Closure. Whether the band-projected densities $\bar{\rho}_{\mathbf{q}} = \mathcal{P}_n e^{-i\mathbf{q}\cdot\mathbf{r}} \mathcal{P}_n$ with the projector $\mathcal{P}_n$ onto the n -th band as a whole (S84) obey *some* closed algebra at all, i.e., whether $[\bar{\rho}_{\mathbf{q}}, \bar{\rho}_{\mathbf{p}}] = f(\mathbf{q}, \mathbf{p}) \bar{\rho}_{\mathbf{q}+\mathbf{p}}$ for a structure constant f [133]. Evaluating the commutator leaves a coefficient $f(\mathbf{q}, \mathbf{p})_{\mathbf{k}}$ that generically still explicitly depends on $\mathbf{k}$ (varying across the BZ), while the right-hand side carries a single coefficient per momentum transfer. Closure therefore requires this residual $\mathbf{k}$ -dependence to cancel, $f(\mathbf{q}, \mathbf{p})_{\mathbf{k}} = f(\mathbf{q}, \mathbf{p})$ [133], so that the structure constant depends on the transferred momenta alone.

GMP form. The Girvin-MacDonald-Platzman (GMP) algebra [131] of the LLL is *one* such closed algebra, but it carries a free function, the form factor $F(\mathbf{q})$, entering as $f \propto [F(\mathbf{q})F(\mathbf{p})/F(\mathbf{q}+\mathbf{p})] \sin[\frac{1}{2}\Omega_{xy}(\mathbf{q} \times \mathbf{p})_z]$, where

the constant Ω_{xy} appearing here is fixed by the Berry curvature [133]. Different form factors describe genuinely different closed algebras, among them those of the higher Landau levels. These share the uniform Berry curvature of the LLL, but obey $\text{tr} g_n = (2n+1)|\Omega_{xy}|$ [93] and hence violate the trace condition below for $n \geq 1$, so that closure alone does not single out the LLL.

Ideal-band condition. A pointwise saturated trace condition $\text{tr} g(\mathbf{k}) = |\Omega_{xy}(\mathbf{k})|$ (S75), which is the property we test here and which, following Ref. [91], we call an ideal, or vortexable, band throughout. Saturation alone already fixes the metric completely in terms of the curvature: combining it with the determinant inequality $2\sqrt{\det g} \geq |\Omega_{xy}|$ (S74) and $2\sqrt{\det g} \leq \text{tr} g$ forces $g_{\mu\nu} = \frac{1}{2}|\Omega_{xy}|\delta_{\mu\nu}$. It does *not*, however, force the curvature itself to be uniform.

LLL-mimicry condition. A uniform Berry curvature $\Omega_{xy}(\mathbf{k}) = \text{const.}$ *in addition* to the saturated trace condition. This is the strictly stronger requirement, and the two must be kept apart: the quarter-flux ground band nearly saturates the trace condition, missing it by 8.3% of $2\pi|C_1|$, while its Berry curvature remains non-uniform.

There are two results which relate them. Roy [58] showed that the LLL-mimicry condition, i.e., saturation *together with* uniform curvature, is necessary and sufficient for closure *with the form factors of the LLL*. Recently, Wang and Simon [133] then showed that this exhausts the possibilities, namely, in two and three dimensions, any closed algebra of band-projected density operators must be of GMP form for *some* form factor, and closure in turn forces a $\mathbf{k}$ -independent QGT. However, the opposite implication is *not* true, i.e., a band with constant Berry curvature and constant quantum metric need not lead to the closure of the projected density algebra [133]. The ideal-band condition is therefore the sharp criterion for the LLL member of the GMP family, not for closure in general.

A. Momentum-space hierarchy: from local positivity to integrated conditions

Every momentum-space criterion stems from a single algebraic fact, namely that the quantum geometric tensor (QGT) is positive semi-definite [59, 60]. For an isolated Bloch band $|\Psi_n(\mathbf{k})\rangle$, let $P_{\perp}(\mathbf{k}) = \mathbb{1} - |\Psi_n(\mathbf{k})\rangle\langle\Psi_n(\mathbf{k})|$ denote the projector onto the orthogonal complement of that band at fixed $\mathbf{k}$, so that the QGT (S35) reads $Q_{\mu\nu}^n(\mathbf{k}) = \langle\partial_{\mu}\Psi_n|P_{\perp}(\mathbf{k})|\partial_{\nu}\Psi_n\rangle = g_{\mu\nu}^n(\mathbf{k}) - \frac{i}{2}\Omega_{\mu\nu}^n(\mathbf{k})$. Inserting a resolution of the identity on the orthogonal complement, $P_{\perp}(\mathbf{k}) = \sum_{m \neq n} |\Psi_m(\mathbf{k})\rangle\langle\Psi_m(\mathbf{k})|$ (a sum over all other bands), gives

$$Q_{\mu\nu}^n(\mathbf{k}) = \sum_{m \neq n} (w_m^{\mu})^* w_m^{\nu}, \quad w_m^{\mu} \equiv \langle\Psi_m|\partial_{\mu}\Psi_n\rangle, \quad (\text{S73})$$

which is a Gram matrix $Q^n = W^\dagger W$ with $(W)_{m\nu} \equiv w_m^\nu$ and is therefore Hermitian and positive semi-definite, denoted $Q^n \succeq 0$, independent of any model details. In the remainder of this section we suppress the band index n wherever the statement holds for an arbitrary isolated band.

Local inequalities. Writing $Q = g - \frac{i}{2}\Omega$ with the antisymmetric $\Omega_{\mu\nu} = \Omega_{xy}\varepsilon_{\mu\nu}$ in two dimensions, the semi-definite positivity of the 2×2 Hermitian matrix Q is equivalent to a non-negative determinant, $\det Q = \det g(\mathbf{k}) - \frac{1}{4}\Omega_{xy}(\mathbf{k})^2 \geq 0$. This yields the so-called *local determinant inequality* [59, 60]

$$2\sqrt{\det g(\mathbf{k})} \geq |\Omega_{xy}(\mathbf{k})|, \quad \forall \mathbf{k}. \quad (\text{S74})$$

Applying the arithmetic-geometric-mean inequality to the non-negative eigenvalues $\lambda_{1,2}$ of the symmetric metric g , $\text{tr } g = \lambda_1 + \lambda_2 \geq 2\sqrt{\lambda_1\lambda_2} = 2\sqrt{\det g}$, and combining it with (S74) yields the so-called *local trace inequality* [59, 60]

$$\text{tr } g(\mathbf{k}) \geq |\Omega_{xy}(\mathbf{k})|, \quad \forall \mathbf{k}. \quad (\text{S75})$$

In total, we thus have the hierarchy of local inequalities

$$\text{tr } g(\mathbf{k}) \geq 2\sqrt{\det g(\mathbf{k})} \geq |\Omega_{xy}(\mathbf{k})|, \quad (\text{S76})$$

which are pointwise statements, valid for any (isolated) band, with no reference to the global topology of the bands. These inequalities are also valid for a filled band with a projector $P(\mathbf{k})$ of rank > 1 [59, 60].

Integrated (global) conditions. Integrating (S75) over the first (magnetic) Brillouin zone (BZ) and using $\int_{\text{BZ}} |\Omega_{xy}| d^2k \geq |\int_{\text{BZ}} \Omega_{xy} d^2k| = 2\pi|C|$ gives the (global) *trace condition*

$$\int_{\text{BZ}} \text{tr } g(\mathbf{k}) d^2k \geq 2\pi|C|, \quad (\text{S77})$$

originally proposed (on LLL-mimicry grounds) by Refs. [79, 90], with C being the first Chern number. Note that the defect of (S77) is in general *not* equal to the integrated local defect $\mathcal{T}_{\text{HH}}$ (9) we introduced the main text, but they agree whenever the Berry curvature does not change sign over the BZ. Only $\mathcal{T}_{\text{HH}}$ is built from a pointwise non-negative integrand $D(\mathbf{k}) = \text{tr } g(\mathbf{k}) - |\Omega_{xy}(\mathbf{k})| \geq 0$, so that it is zero exactly when the band is locally ideal, whereas the defect of (S77) $\int \text{tr } g d^2k - 2\pi|C| \geq 0$ can be non-zero even for a locally ideal band whose curvature changes sign.

Integrating (S74) in the same way, $2 \int_{\text{BZ}} \sqrt{\det g} d^2k \geq \int_{\text{BZ}} |\Omega_{xy}| d^2k \geq 2\pi|C|$, gives the global *determinant condition* [59, 60]

$$\text{vol}_g \equiv \int_{\text{BZ}} \sqrt{\det g} d^2k \geq \pi|C| \quad (\text{S78})$$

A band that in addition carries a $\mathbf{k}$ -independent Berry curvature (*Berry-curvature flatness*) and a $\mathbf{k}$ -independent quantum metric (*metric flatness*), and that

saturates (S77), is band-geometrically a LLL on the lattice [91], since the LLL admits $g_{\mu\nu} = \frac{1}{2}\ell_B^2 \delta_{\mu\nu}$ and $\Omega_{xy} = -\ell_B^2$ with $\ell_B = |B|^{-1/2}$ the magnetic length [60], saturating (S75) pointwise.

Curvature flatness together with saturation of (S77) is Roy's ideal-band condition [58], equivalent to closure of the projected density algebra with LLL form factors. A $\mathbf{k}$ -independent QGT is a weaker requirement than the ideal-band condition. Closure with *any* form factor already enforces $\mathbf{k}$ -independence of the QGT [133], which is therefore necessary for closure, but sufficient neither for closure nor for ideality.

B. Application to the quarter-flux Harper-Hofstadter ground band

We now evaluate every level of the hierarchy on the exact ground band $|\Psi_{-,1}\rangle$, using the closed-form Berry curvature (S44) and quantum metric (S52)–(S54). One feature simplifies our following analysis. Since $\Omega_{xy}^1 = -\sqrt{2}a^2(12 - Z)/Z^{3/2} < 0$ everywhere ($Z \in [4, 8] < 12$), the curvature never changes sign, so that $\int_{\text{BZ}} |\Omega_{xy}^1| d^2k = 2\pi|C_1| = 2\pi$ exactly with $C_1 = -1$. Thus, the integrated defect of (S77) is local in origin and not a curvature sign-cancellation effect, and hence equal to the integrated local defect $\mathcal{T}_{\text{HH}}$ (9) of the main text.

We collect the results in Table S1. Although the band is not flat-geometric ideal with both the Berry curvature and the trace of the metric fluctuating, the *integrated* conditions fall short of saturation only modestly. In particular, the determinant condition (S78) by 4.74% and the trace condition by $\mathcal{T}_{\text{HH}} = 0.522$ (S77), i.e. 8.31%.

As expected the determinant condition is tighter because the arithmetic-geometric-mean step separating (S74) from (S75) costs

$$\mathcal{D}_{\text{ani}}^1(\mathbf{k}) \equiv \text{tr } g^1 - 2\sqrt{\det g^1} = (\sqrt{\lambda_1} - \sqrt{\lambda_2})^2 \geq 0 \quad (\text{S79})$$

with $\lambda_{1,2}$ the eigenvalues of g^1 , so that it vanishes where the metric is isotropic and grows with its anisotropy. Here it vanishes exactly at the C_4 -equivalent momenta Γ , X' , Y' , M' and Σ' , where C_4 enforces $g_{xx} = g_{yy}$ and $g_{xy} = 0$, and reaches its maximum $\mathcal{D}_{\text{ani}}^1 = [\frac{5}{4} - \frac{\sqrt{3}}{3} - \sqrt{\frac{3}{2} - \frac{2}{\sqrt{3}}}]a^2 \approx 0.0850a^2$ at $\mathbf{k}a = (\pi/4, 0)$ and its symmetry-related images [Figs. S2 and S4].

Combining the exact metric trace (S55) with the curvature (S44), written as $|\Omega_{xy}^1|/a^2 = \sqrt{2}(12 - Z)/Z^{3/2}$ (using $6 - c_{4x} - c_{4y} = 12 - Z$, positive since $Z \leq 8$), the local trace-condition defect $D^1(\mathbf{k}) = \text{tr } g^1(\mathbf{k}) - |\Omega_{xy}^1(\mathbf{k})|$

ID	Criterion	Defect measure	HH value
L1	local determinant (S74)	$\mathcal{D}_{\text{det}} = 2\sqrt{\det g} - \Omega_{xy} \geq 0$	mean = 0.0302, max = $\frac{3}{2} - \sqrt{2} \approx 0.0858$
L1'	anisotropy step of (S76)	$\mathcal{D}_{\text{ani}} = \text{tr } g - 2\sqrt{\det g} \geq 0$	mean = 0.0227, max ≈ 0.0850
L2	local trace (S75)	$\text{tr } g - \Omega_{xy} \geq 0$	mean = 0.0529, max = $\frac{5}{4} - \frac{2}{\sqrt{3}} \approx 0.0953$
G1	curvature flatness	$(\Omega_{\text{max}} - \Omega_{\text{min}})/ \bar{\Omega} $	$\pi(4\sqrt{2} - 1)/8 \approx 182.9\%$ ($\Omega \in [-\sqrt{2}, -\frac{1}{4}]$)
G2	metric flatness	$(\text{tr } g_{\text{max}} - \text{tr } g_{\text{min}})/\text{tr } g$	$\frac{5}{4}\pi^2/(2\pi + \mathcal{T}_{\text{HH}}) \approx 181.3\%$ ($\text{tr } g \in [\frac{1}{4}, \frac{3}{2}]$)
G3	determinant condition (S78)	$\int \sqrt{\det g} d^2k - \pi C $	0.1488 (4.74%)
G4	trace condition (S77)	$\mathcal{T}_{\text{HH}} = \int \text{tr } g d^2k - 2\pi C $	0.5220 (8.31%)

TABLE S1. Momentum-space FCI-stability criteria evaluated for the quarter-flux Harper-Hofstadter ground band $|\Psi_{-,1}\rangle$ ($C_1 = -1$). All values in units of the lattice constant $a = 1$, with overbars denoting BZ averages. The two flatness ratios of the metric and curvature are closed-form. With the magnetic BZ area π^2/a^2 one has $\bar{\Omega}_{xy}^1 = 2\pi C_1 a^2/\pi^2 = -2a^2/\pi$ and $\text{tr } g^1 = (2\pi|C_1| + \mathcal{T}_{\text{HH}})a^2/\pi^2$, so that the ranges $\Omega_{xy}^1 \in [-\sqrt{2}, -\frac{1}{4}]a^2$ and $\text{tr } g^1 \in [\frac{1}{4}, \frac{3}{2}]a^2$ giving (G1) = $\pi(4\sqrt{2} - 1)/8$ and (G2) = $\frac{5}{4}\pi^2/(2\pi + \mathcal{T}_{\text{HH}})$ for the curvature and metric flatness, respectively. The BZ integrals are compared against $2\pi|C_1|$ (G4) and $\pi|C_1|$ (G3). The means of L1 and L1' add up to that of L2 by (S82), whereas their maxima do not, since the two defects peak at different momenta. All entries are reproduced from the exact analytic expressions with independent numerical integration over the FBZ.

is obtained in closed form,

$$\frac{D^1(\mathbf{k})}{a^2} = \frac{Z(10 - Z) - 2 + c_{4x}^2 + c_{4y}^2}{Z^2} + \frac{s_{2y}^2 c_{2x}^2 + s_{2x}^2 c_{2y}^2}{(2\sqrt{2Z} + Z)^2} - \frac{\sqrt{2}(12 - Z)}{Z^{3/2}} \geq 0, \quad (\text{S80})$$

which has been depicted in Fig. 2 of the main text.

The two steps of the local hierarchy (S76) resolve D^1 into two separately non-negative pieces. Alongside the anisotropy defect (S79) we define the determinant-condition defect

$$\mathcal{D}_{\text{det}}^1(\mathbf{k}) \equiv 2\sqrt{\det g^1} - |\Omega_{xy}^1| \geq 0, \quad (\text{S81})$$

which is non-negative by (S74), so that the local trace defect splits exactly and pointwise as

$$D^1(\mathbf{k}) = \mathcal{D}_{\text{ani}}^1(\mathbf{k}) + \mathcal{D}_{\text{det}}^1(\mathbf{k}). \quad (\text{S82})$$

Both terms are shown over the FBZ in Fig. S4, together with $\det g^1$ (S56) and $\text{tr } g^1$ (S55) themselves. Since Ω_{xy}^1 does not change sign, we again have $\int_{\text{BZ}} |\Omega_{xy}^1| d^2k = 2\pi|C_1|$, so that the second term gives twice the defect of the global determinant condition (S78), $\mathcal{T}_{\text{HH}} = \int_{\text{BZ}} \mathcal{D}_{\text{ani}}^1(\mathbf{k}) d^2k + 2[\text{vol}_{g^1} - \pi|C_1|]$, making it not an independent measure, but the sum of the metric anisotropy and twice the determinant-condition defect G3 of Table S1.

At the Γ -point and its C_4 -symmetric images X', Y', M' ($Z = 8$) one finds $\text{tr } g^1 = 2\sqrt{\det g^1} = |\Omega_{xy}^1| = a^2/4$, with $\det g^1 = a^4/64$ where the entire chain (S76) therefore collapses to equalities, so that both local inequalities saturate exactly, and the band looks locally like an LLL. However, at the band-edge point Σ' with $\mathbf{k}a = (\pi/4, \pi/4)$ ($Z = 4$) the trace defect is $\text{tr } g^1 - |\Omega_{xy}^1| = \frac{3}{2}a^2 - \sqrt{2}a^2 \approx 0.0858a^2 > 0$, and the global maximum of the local defect, $D^1 = (\frac{5}{4} - \frac{2}{\sqrt{3}})a^2 \approx 0.0953a^2$, is located on the Γ - X' line (and its images). Since the metric is isotropic

at Σ' , the anisotropy defect (S79) vanishes there, so that there the whole trace defect is of determinant type, $\mathcal{D}_{\text{det}}^1 = \frac{3}{2}a^2 - \sqrt{2}a^2$, which at the same time the global maximum of $\mathcal{D}_{\text{det}}^1$ over the FBZ [Fig. S4(c)].

Finally, let us note how the sublattice decomposition $g^1 = \frac{1}{2}(g^{(u_1)} + g^{(v_1)}) + \frac{1}{4}|\Delta A^1|^2$ (S38) resolves the origin of the defect. Since $\Omega_{xy}^{(u_1)}, \Omega_{xy}^{(v_1)} \leq 0$ everywhere and never cancel, the trace defect splits cleanly,

$$\mathcal{T}_{\text{HH}} = \frac{1}{2}\mathcal{T}^{(u)} + \frac{1}{2}\mathcal{T}^{(v)} + \frac{1}{4}\int_{\text{BZ}} |\Delta A^1|^2 d^2k, \quad (\text{S83})$$

with each sector defect $\mathcal{T}^{(u/v)} = \int_{\text{BZ}} [\text{tr } g^{(u/v)} - |\Omega_{xy}^{(u/v)}|] \geq 0$ guaranteed to be non-negative by the PSD applied to the QGT $Q_{\mu\nu}^{u_j/v_j}$ (S36) from the u/v sectors. We find that $\mathcal{T}^{(u)} = \mathcal{T}^{(v)} \approx 0.4917$ and $\int |\Delta A^1|^2 d^2k \approx 0.1211$, reproducing $\mathcal{T}_{\text{HH}} = 0.522$ such that the Berry-connection-difference term, which is absent for a genuine single-orbital LLL, contributes $\frac{1}{4}\int |\Delta A^1|^2/\mathcal{T}_{\text{HH}} \approx 5.8\%$ of the total defect on its own, identifying the internal u/v structure of the system as an additional intrinsic, gauge-invariant obstruction to ideality.

C. Relation to the projected density algebra

Let us briefly connect the band-geometric criteria above to the language of projected (number) density operators. The Fourier transform of the real-space number density operator is $n_{\mathbf{q}} = e^{-i\mathbf{q}\cdot\mathbf{r}}$, which transfers momentum $\mathbf{q}$ to the particle. Below the band gap the particle cannot leave the band, so that every operator of the effective low-energy theory appears in its projected form. For instance a density-density interaction becomes $\sim \sum_{\mathbf{q}} V(\mathbf{q}) \bar{\rho}_{\mathbf{q}} \bar{\rho}_{-\mathbf{q}}$, where the algebra of the projected density $\bar{\rho}_{\mathbf{q}}$ controls the low-energy physics [58, 90, 133]. Since the density operator transfers momentum, the relevant projector here is not the $\mathbf{k}$ -resolved $P_n(\mathbf{k}) =$

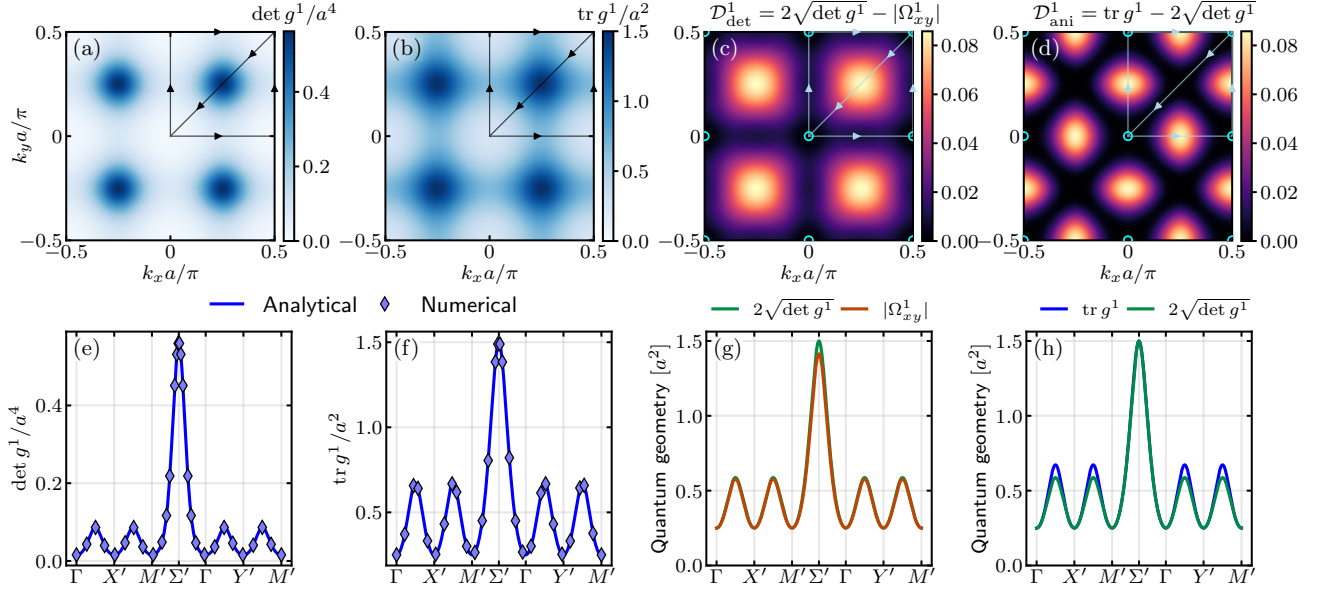


FIG. S4. Local FCI-stability inequalities (S76) evaluated on the lowest (and highest) energy band of the quarter-flux Harper-Hofstadter model, $|\Psi_{\pm,1}\rangle$ (S32). (a)-(d) Analytically obtained determinant $\det g^1$ (S56) and trace $\text{tr } g^1$ (S55) of the quantum metric, and the two local defects $\mathcal{D}_{\text{det}}^1$ (S81) and $\mathcal{D}_{\text{ani}}^1$ (S79), as function of $\mathbf{k}$ over the FBZ. The two contributions to the trace-condition defect $D^1 = \mathcal{D}_{\text{ani}}^1 + \mathcal{D}_{\text{det}}^1$ (S82) [cf. Fig. 2 in main text] become clear. Both defects vanish at the $Z = 8$ momenta Γ, X', Y', M' (cyan circles), where the band is locally an ideal LLL. (e)-(h) Same as (a)-(d) but along the high-symmetry path $\Gamma-X'-M'-\Sigma'-\Gamma-Y'-M'$ (as indicated via solid lines with arrows in (a)-(d), see also Fig. 1 in the main text).

$|\Psi_n(\mathbf{k})\rangle\langle\Psi_n(\mathbf{k})|$ used for the QGT above, but the projector onto the band as a whole,

$$\mathcal{P}_n = \int_{\text{BZ}} d^2k |\Psi_{n,\mathbf{k}}^B\rangle\langle\Psi_{n,\mathbf{k}}^B|, \quad |\Psi_{n,\mathbf{k}}^B\rangle = e^{i\mathbf{k}\cdot\mathbf{r}}|\Psi_n(\mathbf{k})\rangle, \quad (\text{S84})$$

where throughout this work $|\Psi_n(\mathbf{k})\rangle$ denotes the cell-periodic Bloch function, i.e., the eigenvector of $H(\mathbf{k})$ (2), and $|\Psi_{n,\mathbf{k}}^B\rangle$ the full Bloch state carrying the plane-wave factor [175]. The band-projected number density operator is then $\bar{\rho}_{\mathbf{q}} = \mathcal{P}_n n_{\mathbf{q}} \mathcal{P}_n = \mathcal{P}_n e^{-i\mathbf{q}\cdot\mathbf{r}} \mathcal{P}_n$, which is a single-particle operator. Inserting (S84) twice then gives,

$$\bar{\rho}_{\mathbf{q}} = \int_{\text{BZ}} d^2k \langle\Psi_n(\mathbf{k})|\Psi_n(\mathbf{k}+\mathbf{q})\rangle |\Psi_{n,\mathbf{k}}^B\rangle\langle\Psi_{n,\mathbf{k}+\mathbf{q}}^B|. \quad (\text{S85})$$

We see that the plane waves have enforced momentum conservation and the overlap of the cell-periodic Bloch functions $|\Psi_{n,\mathbf{k}}^B\rangle\langle\Psi_{n,\mathbf{k}+\mathbf{q}}^B|$ survives as the weight. Expanding that overlap to second order in $\mathbf{q}$ returns the QGT, which is why the quantum geometry of Section S6 A controls the algebra [176]

The projection is what makes this algebra non-trivial. Namely, while unprojected densities multiply as $n_{\mathbf{q}}n_{\mathbf{p}} = n_{\mathbf{q}+\mathbf{p}}$ and hence commute, $\bar{\rho}_{\mathbf{q}}\bar{\rho}_{\mathbf{p}}$ do not. Acting on a Bloch state at $\mathbf{k}$, its leading order in the momenta gives

the Berry curvature [90],

$$[\bar{\rho}_{\mathbf{q}_1}, \bar{\rho}_{\mathbf{q}_2}]|\Psi_{n,\mathbf{k}}^B\rangle = -i(\mathbf{q}_1 \times \mathbf{q}_2)_z \Omega_{xy}^n(\mathbf{k}) \bar{\rho}_{\mathbf{q}_1+\mathbf{q}_2}|\Psi_{n,\mathbf{k}}^B\rangle + \mathcal{O}(q^3). \quad (\text{S86})$$

We immediately see that closure already at this order would demand a constant Berry curvature [90] while closure at all wavelengths, with LLL form factors, additionally demands pointwise saturation of the trace condition [58, 91]. Both of these requirements are not satisfied here as detailed in the previous section Section S6 B. This is not specific to the quarter-flux model, since a closed band-projected density algebra cannot be realized exactly in any tight-binding model with finitely many orbitals per unit cell [132], as the four-orbital band studied here.

S7. STATIC STRUCTURE FACTOR AND THE QUANTUM WEIGHT

At last, let us now provide some details on the connection between the band geometry and the long-wavelength limits of the static structure factor of a filled band, and its connection to general topological bounds, as recently revealed in [61, 62, 169]. The static structure factor is a directly measurable quantity which can be obtained in scattering experiments, or via optical conductivity measurements through sum rules [61, 62, 169].

The static structure factor is defined for any state, interacting or not, as the equal-time correlator of the number density operator $\rho(\mathbf{r})$ of the many-body ground state,

$$S(\mathbf{q}) = \frac{1}{V} \langle \rho_{\mathbf{q}} \rho_{-\mathbf{q}} \rangle, \quad \rho_{\mathbf{q}} = \int d\mathbf{r} e^{-i\mathbf{q} \cdot \mathbf{r}} \rho(\mathbf{r}), \quad (\text{S87})$$

with V the volume, following the normalization of Ref. [61]. Note that $\rho_{\mathbf{q}}$ here is the *many-body* density operator $\int d\mathbf{r} \hat{\Psi}^\dagger(\mathbf{r}) e^{-i\mathbf{q} \cdot \mathbf{r}} \hat{\Psi}(\mathbf{r})$, the second-quantized image of the one-body operator $e^{-i\mathbf{q} \cdot \mathbf{r}}$. Expanding the field operator $\hat{\Psi}(\mathbf{r})$ in the Bloch basis for non-interacting fermions filling the lowest Hofstadter band, we obtain

$$\rho_{\mathbf{q}} = \sum_{\mathbf{k}} \langle \Psi_{-,1}(\mathbf{k}) | \Psi_{-,1}(\mathbf{k} + \mathbf{q}) \rangle c_{\mathbf{k}}^\dagger c_{\mathbf{k}+\mathbf{q}}. \quad (\text{S88})$$

where $c_{\mathbf{k}}^\dagger$ denotes the creation of a fermion in mode $|\Psi_{-,1}^B(\mathbf{k})\rangle$. The emerging overlap factor $\langle \Psi_{-,1}(\mathbf{k}) | \Psi_{-,1}(\mathbf{k} + \mathbf{q}) \rangle$ is the entire origin of the band geometry in $S(\mathbf{q})$. For the filled Fermi sea of the lowest band, we now insert (S88) into (S87), using $\text{Tr}[P_1(\mathbf{k})P_1(\mathbf{k} + \mathbf{q})] = |\langle \Psi_{-,1}(\mathbf{k}) | \Psi_{-,1}(\mathbf{k} + \mathbf{q}) \rangle|^2$ together with $V^{-1} \sum_{\mathbf{k}} \rightarrow \int d^2k / (2\pi)^2$, and obtain [61],

$$S(\mathbf{q}) = \int_{\text{BZ}} \frac{d^2k}{(2\pi)^2} \text{Tr}[P_1(\mathbf{k})(P_1(\mathbf{k}) - P_1(\mathbf{k} + \mathbf{q}))]. \quad (\text{S89})$$

with the ground-band projector $P_1(\mathbf{k}) = |\Psi_{-,1}(\mathbf{k})\rangle\langle\Psi_{-,1}(\mathbf{k})|$ of Section S6 A. The integrand is the gauge-invariant quantum distance $1 - |\langle \Psi_{-,1}(\mathbf{k}) | \Psi_{-,1}(\mathbf{k} + \mathbf{q}) \rangle|^2 = g_{\mu\nu}^1 q^\mu q^\nu + \mathcal{O}(|q|^3)$, which expanded to second order gives the quantum metric $g_{\mu\nu}^1$ of the lowest band [2]. Thus, the so-called *quantum weight* $K_{\mu\nu}$ [61, 62], is obtained

$$S(\mathbf{q}) = \frac{K_{\mu\nu}}{2\pi} q^\mu q^\nu + \mathcal{O}(q^4), \quad K_{\mu\nu} = \frac{1}{2\pi} \int_{\text{BZ}} d^2k g_{\mu\nu}^1(\mathbf{k}). \quad (\text{S90})$$

The C_4 symmetry of the model makes the quantum weight isotropic, $K_{xx} = K_{yy} \approx 0.5415$ and $K_{xy} = 0$. Because the Berry curvature does not change sign over the BZ, K is not independent geometric data but actually the integrated trace condition of the main text in disguise,

$$K \equiv K_{xx} + K_{yy} = |C_1| + \frac{\mathcal{T}_{\text{HH}}}{2\pi} \approx 1.083 \geq |C_1| = 1, \quad (\text{S91})$$

i.e., the 8.3% by which K exceeds the topological bound $K \geq |C|$ [61] is exactly $\mathcal{T}_{\text{HH}}/2\pi$, so the structure factor reformulates the same geometric defect as a measurable density correlator. Note that, up to the factor 2π , K is also the gauge-invariant part of the Marzari-Vanderbilt Wannier spread functional [39, 42], equivalently the localization functional of Refs. [41, 61]. The bound $K \geq |C|$ is thus equivalent to non-zero Chern number obstructing the existence of exponentially localized Wannier functions [170, 171], where $|C|$ puts a floor on the Wannier spread. For the ground band this floor accounts for $|C_1|/K \approx 92.3\%$ of the quantum weight, the remaining 7.7% being the excess spread that the band carries beyond its topological minimum, which is the same 8.3% excess mentioned above measured relative to the bound $|C_1|$ (and not to K). This excess here is a measure of non-ideality (rather than of delocalization), since a band saturating $K = |C|$ would have the most localized Wannier functions its topology permits, while still decaying algebraically rather than exponentially due to $C \neq 0$ [170, 171].